Forecasting Emerging News Trends using Natural Language Processing (NLP), Network Science and Machine Learning

By

Alan Arimpur Raju

Student ID: 2832910

Supervisor: Dr. Jens Christian Claussen

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"I certify that this project is my own work. Code development and debugging were assisted by ChatGPT-5. The report structure and initial drafts are my own; final editing was assisted by ChatGPT to improve clarity, consistency, and grammatical accuracy. All outputs have been reviewed and edited by me to ensure they accurately reflect my work."

Abstract

This dissertation investigates short-horizon news-trend forecasting by integrating network science and machine learning on a five-week corpus (~190k articles, 16 July 2025–22 August 2025) collected via NewsAPI. Two daily graph views are constructed: (i) entity co-occurrence networks and (ii) article-embedding similarity networks (Sentence Transformers + FAISS). Communities are detected with the Leiden algorithm (cuGraph) and linked across days into longitudinal "global communities." Four centralities (degree, betweenness, closeness, eigenvector) are computed, then combined via PCA into a composite indicator (y).

Two forecasting routes are evaluated. Method 1 (classification) predicts whether a community will trend next day using network features. On entity graphs, models achieve strong discrimination (Random Forest AUROC 0.964, AUPRC 0.812; Logistic Regression highest recall). Embedding graphs add semantic coverage but are less robust to event shocks (best AUROC ≈ 0.89, AUPRC ≈ 0.52). Method 2 (time series) forecasts y per community. A Huber-slope momentum is intuitive but brittle to regime shifts, while Prophet's piecewise-linear trend with uncertainty yields reliable direction signals on entity communities (average sMAPE ≈ 8.3%, coverage ≈ 0.75) but not shock-responsive on embedding communities.

Findings show that structural signals in entity networks provide the most reliable early indicators of virality, while embeddings supply complementary thematic context. The work contributes a scalable pipeline, comparative benchmarks, and a practical framework for surfacing emerging narratives for journalists, businesses, and policymakers. Limitations (English-dominant, five-week span; event surges) motivate future work on multi-year, multilingual data, exogenous signals, and hybrid/event-aware models.

Acknowledgment

I would like to express my deepest gratitude to my supervisor, Dr. Jens Christian Claussen, for his invaluable guidance and unwavering support throughout the course of this dissertation. His insightful feedback, constructive criticism, and constant encouragement not only shaped this work but also inspired me to approach research with greater curiosity and perseverance. Witnessing his passion for science and dedication to inquiry has been highly motivating, and it has been both an honor and a privilege to work under his supervision.

Above all, I am profoundly thankful to the Almighty for being a constant source of strength and light in moments of challenge and uncertainty. This faith has nurtured my confidence, sustained my motivation, and ultimately guided me to the successful completion of this project.

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1. Introduction

In today's digital world, information spreads at unprecedented speed, with news consumption increasingly shaped by online platforms. Social media and digital journalism create dynamic ecosystems where stories emerge, evolve, and disappear within hours. Forecasting which topics will trend offers stakeholders a competitive edge: journalists can prioritise impactful stories, businesses can anticipate consumer needs and market opportunities, and policymakers can detect early signals of social or economic issues to make timely decisions.

Traditional news analysis based on keyword frequency or sentiment often misses the relational structure of narratives. Network science addresses this by mapping entity co-occurrences and relationships, revealing interconnected topic communities. Combined with machine learning, such networks offer predictive insight into how stories gain momentum. However, forecasting remains difficult due to the volume, velocity, and variability of news. This dissertation integrates network science and machine learning to forecast emerging trends, providing a framework for more accurate prediction and highlighting the value of interdisciplinary methods in complex digital ecosystems.

1.1 Problem Statement

With global access expanding, news now spreads at unprecedented volumes and speeds but is often diluted by bias, sensationalism, and clickbait. Recent work shows its value for forecasting economic indicators (Ashwin et al., 2024), motivating systems that can surface emerging topics before they peak. Existing approaches struggle with scale, linguistic diversity, and concept drift, which hinder early and reliable detection. While network-based methods have predicted virality in social platforms (Weng, Menczer & Ahn, 2013; Xu et al., 2023), few systems apply these insights to news. This gap underscores the need for methods that detect and forecast emerging news trends early, accurately, and with interpretable context.

Existing approaches focus mainly on keywords or sentiment, offering descriptive rather than predictive insights. This highlights the need for methods that integrate network analysis with predictive modelling to forecast future trending news communities.

1.2 Research Aim and Objectives

The aim of this dissertation is to develop and evaluate a framework for predicting emerging news trends using communities by applying network analysis and machine learning techniques.

To accomplish this aim, the dissertation pursues the following objectives:

- Construct daily co-occurrence networks of named entities extracted from news articles.
- Construct daily embedding-based similarity networks using cosine similarities (FAISS) of the news articles.
- Apply community detection algorithms to identify and track evolving clusters within both types of networks.
- Compute a range of centrality and structural features (degree, betweenness, closeness, eigenvector centralities) to characterize the identified communities.
- Develop, train, and evaluate machine learning models to forecast the likelihood of communities gaining traction over subsequent days, and compare their performance across network types while critically analysing their relative strengths and limitations.

1.3 Research Questions

This dissertation aims to find answers for the following research questions

- Can community detection and network features provide predictive signals of emerging news trends?
- Which forecasting methods and machine learning models are most effective?
- How can trend-based signals be used effectively?
- What are the practical challenges and limitations of short-term news trend prediction?

1.4 Methodological Overview

The proposed approach begins with preprocessing a dataset of news articles to extract entities and vectorized representations of the news articles then construct daily co-occurrence and embedding based networks respectively. Communities are then detected followed by deriving centrality features from the network which include degree, betweenness, closeness, eigenvector centralities. A set of supervised machine learning models is trained to predict which communities will trend in the near future based on the information that was extracted from the network. The models are evaluated using standard performance metrics.

2. Literature Review

The rapid growth of digital media has created a pressing need to detect and forecast news trends within high-volume information streams. Traditional methods such as keyword indexing and semantic search aid retrieval but fail to capture deeper relational structures (Blokh & Alexandrov, 2015). Network science addresses this by modelling articles or entities as nodes and their semantic, temporal, or social links as edges, enabling the detection of hubs, clusters, and influence pathways that reveal significant or trending stories. Blokh and Alexandrov (2015) showed that Twitter networks exhibit scale-free structures, where hubs correspond to major news events, with retweet counts reflecting societal impact. Similarly, Nicholls and Bright (2018) developed automated methods to identify news story chains through similarity measures and clustering, demonstrating the prevalence of ‘mega stories’ in UK media and confirming that news evolves in interconnected chains rather than isolated articles, underscoring the value of network clustering for analysing news dynamics.

Community detection is central for identifying emerging news clusters. While the Louvain algorithm was once popular for its efficiency, it often produced poorly connected partitions. The Leiden algorithm (Traag, Waltman & Van Eck, 2019) overcomes these issues, ensuring well-connected, higher-quality communities and outperforming Louvain in both speed and accuracy. Therefore making it especially relevant for news trend prediction.

For scalability, this work used RAPIDS cuGraph’s GPU-accelerated implementation of Leiden, which parallelises local move, refinement, and aggregation steps via CUDA kernels. This delivers substantial throughput gains over CPU libraries, enabling near real-time analysis on graphs at billion-edge scale (NVIDIA, 2021; 2025a; Sahu, 2024).

Node centrality was computed using nx-cugraph, which dispatches supported algorithms to RAPIDS cuGraph for GPU acceleration, enabling degree, betweenness, and eigenvector centralities to run efficiently on large graphs (NVIDIA, 2025b). As cuGraph (2025) lacked closeness centrality, NetworkKit was used as a multi-core CPU fallback, leveraging parallelism to speed calculations. This GPU–CPU combination ensured scalability for large-scale forecasting tasks.

In parallel, studies have investigated embedding-based methods and natural language processing methodologies. Velay and Daniel (2018) explored using NLP on financial news headlines to predict stock index trends, testing classifiers such as SVM, Random Forest and LSTMs on sentiment and contextual features. Although accuracy was modest, their study highlights the potential of combining linguistic embeddings with machine learning for trend forecasting.

Beyond structural and clustering approaches, a key line of research focuses on predictive models that leverage network-derived features. Classical models such as logistic regression have long been employed for binary classification tasks in social and communication studies. This includes the detection of whether a news item is likely to gain traction from using news headlines (Velay and Daniel, 2018). Logistic regression offers interpretablity and remains effective when combined with carefully engineered features such as centrality measures derived from temporal networks.

Recent advances in machine learning have introduced gradient boosting frameworks such as LightGBM, designed for efficiency and strong predictive performance on large, high-dimensional data. LightGBM has proven effective in text mining and forecasting by capturing non-linear feature interactions at scale (Ke et al., 2017), and in news trend prediction it can exploit centrality-based features from temporal networks to identify potential breakouts. Time-series models have also been adopted, with Facebook's Prophet gaining popularity for its robustness to irregularity, seasonality, and missing data (Taylor & Letham, 2018). Applied to aggregated features from network analysis, Prophet provides insight into whether a trend is likely to persist or decay over time.

Together, these models represent a methodological spectrum: from interpretable linear and logistic regression approaches, through non-linear ensemble methods such as LightGBM, to dedicated time series forecasters such as Prophet. Their combined application marks the importance of interdisciplinary nature of news trend prediction, bridging network science feature engineering with machine learning and statistical forecasting.

Accurate representation of textual content is a prerequisite for effective news trend prediction, particularly when linking entities or clustering articles into evolving stores. Traditional approaches have relied on cosine similarity applied to TF-IDF or bag-of-words vectors, which despite their simplicity, have been widely used in early studies of document retrieval and clustering (Manning, Ragavan and Schütze, 2008). However, such methods often fail to capture deeper semantic relations between entities or across news narratives.

Recent advances in transformer-based language models have significantly improved embedding quality by encoding contextual and semantic information, The all-mpnet-base-v2-model, developed within the Sentence Transformer framework, provides dense, semantically meaningful embeddings optimized for similarity and clustering tasks (Song et al., 2020). When applied to news articles or extracted named entities, these embeddings enable more reliable detection of semantic overlap than surface level word matching, thereby enhancing both entity co-occurrence networks and article similarity graphs.

The scale of modern news corpora demands efficient similarity search methods. FAISS, developed by Facebook AI Research, enables fast approximate nearest neighbour search in high-dimensional spaces (Johnson, Douze & Jégou, 2019). Combined with transformer embeddings, it rapidly identifies semantically similar articles across millions of records, making it ideal for large-scale trend analysis. Using transformer embeddings with FAISS and cosine similarity provides state-of-the-art semantic matching, ensuring communities are grouped not just by text but by meaning, strengthening downstream tasks such as community detection, centrality analysis, and predictive modelling.

3. Methodology

This chapter outlines the methodological framework used in this research. Covering the data collection, preprocessing, network creation, feature engineering, modelling approaches, evaluation metrics and experimental setup.

3.1 Data Collection

The news dataset was obtained from NewsAPI (newsapi.org), a widely used RESTful aggregation service providing JSON-formatted results from over 150,000 sources worldwide. Using the free Developer tier, which limits access to 100 requests per day (10,000 news per day) and articles from the past 30 days, daily requests were issued at maximum capacity between 16 July 2025 and 22 August 2025, yielding about 190,000 articles for analysis. This method ensured the collection of the latest real world news analysis.

The raw dataset comprised of numerous attributes, of which the following were retained as relevant to the scope of this thesis:

- Topic: The search query or topical classification guiding article retrieval.
- Source Name: The provenance of each news article (eg., BBC, Reuters).
- Title: the headline text of the article.
- Description: A concise summary longer than the title, yet shorter than the full content.
- PublishedAt: the timestamp denoting when the article was released publicly.

API attributes such as URLs, images, and author names were excluded as irrelevant. While authorship could indicate influence or bias, the dataset's limited scope and low variability made it uninformative, and missing values further reduced its usefulness.

3.2 Data Preprocessing

Preprocessing was crucial to ensure dataset integrity, removing issues like inconsistent encodings, duplicates, and multilingual content that would otherwise compromise reliable feature extraction and prediction (Han, Kamber and Pei, 2011).

3.2.1 Text Normalization

A multi-step normalisation pipeline was applied to the concatenated title and description fields to create a consistent clean_text column, a crucial stage in NLP for reducing noise, lexical sparsity, and ensuring token comparability (Manning, Raghavan & Schütze, 2008; Jurafsky & Martin, 2023). Unicode canonicalisation (unicodedata.normalize('NFC')) unified composite and accented characters, while regex substitutions standardised punctuation, removed unwanted characters, URLs, and emails. The cleantext library further eliminated emojis and residual web tokens, producing linguistically stable text free from mis-encodings.

A stricter lexical filtering stage then lowercased tokens, removed mentions, stripped non-alphabetic characters, discarded tokens under three characters, and applied stopword removal (Fox, 1989). This yielded a vocabulary of content-bearing tokens suitable for semantic analysis. By combining these steps: Unicode normalisation, regex cleaning, pruning, and stopword removal, the final clean_text column ensured that co-occurrence and embedding networks, as well as centrality features, were built on clean, consistent, and reliable text (Camacho-Collados & Pilehvar, 2018; van der Goot et al., 2020).

3.2.2 Language Detection

Since the dataset spanned multiple languages, a fastText-based language identification model (lid.176.bin) was applied to each text entry (Joulin et al., 2017). This enabled the filtering of documents to those in English, ensuring homogeneity of linguistic features and comparability of results. Automated language identification is a recognised best practice in multilingual corpora to prevent spurious feature distributions across languages (Huang and Lee, 2022).

A histogram of detected languages (*Figure B.1*) shows the dataset is dominated by English ($>10^5$ articles), with Spanish, German, Portuguese, and French in much smaller numbers, and only a few cases of languages like Chinese and Thai. This confirmed the corpus's multilingual nature and justified restricting analysis to English for consistency and comparability.

3.2.3 Deduplication

Large-scale news corpora often contain duplicates from syndication and reposting, which inflate term frequencies, bias community detection, and cause overfitting (Elmagarmid, Ipeirotis & Verykios, 2007; Papadakis et al., 2021). To address this, a two-stage deduplication strategy was applied: blocking with a normalised hash of the first 40 title characters to form candidate sets (Christen, 2012), followed by TF-IDF vectorisation (unigrams–bigrams, 10,000 features) and cosine similarity filtering, where pairs above 0.90 were removed. This enabled efficient large-scale deduplication while preserving content diversity (Ramos, 2003; Manning, Raghavan & Schütze, 2008).

This pipeline ensured that only a single representative was retained for higher similar items, balancing efficiency and accuracy while reducing false positives through conservative thresholds and lexical blocking.

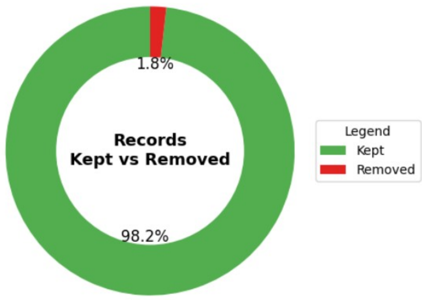

Figure 1 : Duplicate removal percentage

Of 186,736 articles (post-normalisation), 3,356 duplicates (1.8%) were removed, leaving 183,380 unique items. Though small, this pruning improved dataset diversity and reduced bias from redundant content (Brodley and Friedl, 1999).

3.2.4 Processing Findings and Analysis

The preprocessed corpus was examined in terms of temporal coverage, source composition, topical distribution and category dynamics. Such exploratory profiling not only highlights the scale and diversity of the dataset but also reveals potential biases, such as source concentration or class imbalance that may influence predictive performance.

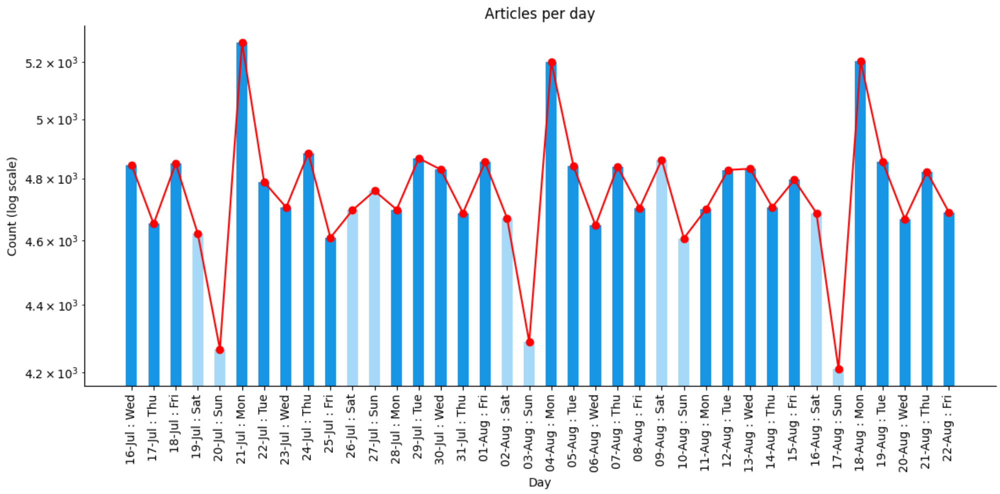

Figure 2 : Temporal Coverage

The dataset displayed (*Figure 2*) relatively stable daily article counts across the collection period (16 July – 22 August), with volumes ranging from approximately 4,200 to 5,200 articles per day (*Figure 2*). Peaks were observed on Mondays, reflecting weekly reporting cycles where news output typically increases following weekends (Leskovec, Backstrom and Kleinberg, 2009). The stability of article counts ensures that downstream models are not disproportionately influenced by temporal sparsity or anomalies.

As per *Figure B.2*, the majority of articles originated from a relatively small set of high-volume outlets, with ETF Daily News, Biztoc.com, and Yahoo Entertainment contributing the most content (*Figure B.2*). Despite this concentration, the dataset exhibited broad coverage across topical categories, including business, technology, finance, and science. While the dominance of a few aggregators reflects real-world news distribution patterns, the diversity of sources supports representativeness across domains.

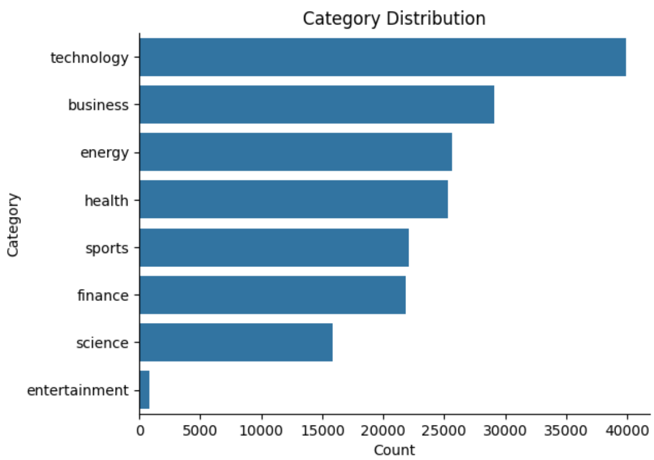

Figure 3 : Topical Category Distribution

Figure 3 shows technology (~40k) as the most prevalent category, followed by business (~29k), energy (~25k), and health (~25k). Sports, finance, and science were smaller, while entertainment was under-represented. This uneven distribution reflects media priorities but also poses class imbalance challenges for supervised models (He & Garcia, 2009).

Disaggregated by category (*Figure B.3*), daily article volumes showed consistent coverage across business, technology, finance, and health. Despite fluctuations, proportions remained stable, confirming trends were not artefacts of bias or short-term skews. Post-normalisation, the dataset spans millions of words across diverse domains, making it well-suited for network analysis and forecasting, though safeguards like class balancing remain necessary.

3.3 Network Construction (*Network Images shown in Figure C.8 & C.9 in appendix C*)

3.3.1 Co-Occurrence Network Creation

A co-occurrence network was built by combining NLP with graph-based modelling to detect potential news trends. Named Entity Recognition (NER) using a transformer-based spaCy model extracted entities such as people, organisations, and locations from the preprocessed corpus. NER was chosen for its ability to capture semantically meaningful units central to news narratives and indicative of emerging trends (Marrero et al., 2013; Jurafsky & Martin, 2023). Entities were then deduplicated within each document to avoid redundancy.

Short Name	Full Name	Description
ORG	Organization	Names of companies, agents, institutions. etc
PERSON	Person	Individual people, includes real or fictional.
GPE	Geopolitical Entity	Names of countries, cities, states and regions.
NORP	Nationalities/Religions/ Political Groups	Names of groups
PRODUCT	Product	Names of commercial products.

Table : Representing targetted labels for entities

The distribution of named entities reveals that news coverage is dominated by organisations, individuals, and countries, reflecting the central role of these actors in shaping narratives. As seen in *Figure B.4*

An undirected graph was then built with entities as nodes and co-occurrences as weighted edges, emphasising frequent associations. This structure suits trend prediction, as recurrent co-occurrences often indicate clusters of attention likely to escalate into trends (Sayyadi et al., 2009).

Average Number of Nodes (with time)	7237
Average Number of Edges (with time)	18918

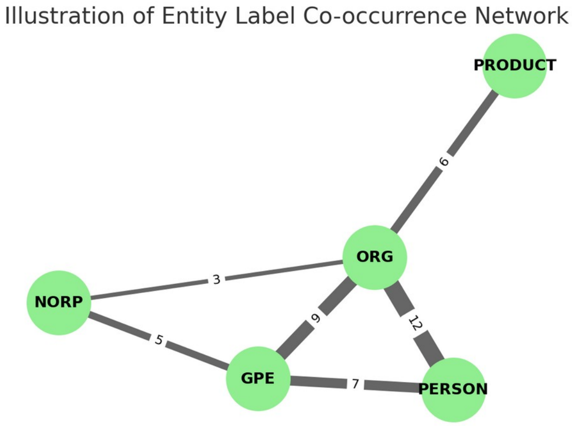

Figure 4 : Illustration of network modelling

Network analysis methods, including centrality and community detection, can then be applied to this structure to identify influential entities, uncover hidden associations, and detect emerging communities of discourse (Barabási, 2016; Newman, 2018). Thus, the co-occurrence network serves as a foundation for mapping the evolving relational structure of news content and provides a systematic way to detect patterns that are predictive of trends.

Average Number of Nodes (with time)	2317
Average Number of Edges (with time)	8052

3.3.2 Embedding-Based Network Creation

The embedding-based network was built using 768-dimensional sentence embeddings, representing the preprocessed news texts in a semantic space. Unlike the co-occurrence network, which captures surface associations, this approach incorporates contextual meaning for a more nuanced analysis.

Embeddings were generated using the pre-trained transformer model all-mpnet-base-v2 within the SentenceTransformers framework (Reimers & Gurevych, 2019), chosen for its strong performance in capturing semantic similarity (Wang et al., 2020). Each news document was encoded into a dense vector, and pairwise similarities were then computed.

To handle large-scale similarity efficiently, FAISS (Facebook AI Similarity Search) with GPU acceleration was used (Johnson et al., 2019), enabling fast nearest-neighbour searches and adaptive thresholding. We then constructed a network where each node is an article, and edges connect documents exceeding the dynamically determined similarity threshold.

The threshold was determined using the elbow method: similarity scores were first normalized (minmax) then sorted, and a straight line drawn between the highest and lowest points of the curve. Distances from this line were calculated (as shown in *Figure 5*), and the point with maximum distance was selected as the elbow, which was then used as the threshold for network construction.

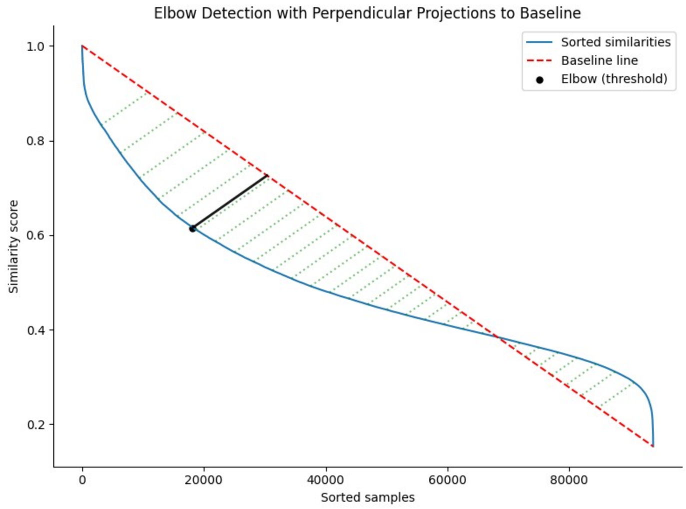

Figure 5 : Visual for threshold calculation for one of the time windows.

This approach ensures that the network retains only the strongest semantic associations, reducing noise while preserving meaningful contextual links between articles. Nodes in the resulting graph represent individual articles, while weighted edges capture their degree of semantic relatedness.

Cosine Similarity	**FAISS**
Compares all vectors directly	Finds nearest vectors and then compare vectors
Slow on large data	Much faster than traditional cosine similarity
Very memory intensive for large datasets	More memory efficient, handles large vectors.

Table : Comparison between Cosine Similarity and FAISS

A sentence embedding–based network was used for trend prediction due to its ability to capture semantic and contextual alignment beyond keyword overlap. This is crucial in news analysis, where different articles may describe the same event. By linking semantically similar content, the network reveals discourse clusters that signal emerging topics or shifts in media attention (Cai et al., 2022), complementing the co-occurrence network by uncovering higher-level narrative patterns for forecasting.

3.3.3 Community Detection and Global Community

Community detection using Leiden algorithm (GPU boosted) was applied to daily networks from both approaches to group related articles/entities into topical "communities of discourse" (Fortunato, 2010). This step is crucial for trend prediction, as rises, persistence, or declines in communities reflect the emergence, diffusion, or fading of news stories (Rosvall & Bergstrom, 2008; Newman, 2006).

CPM Quality Score Average for Co-occurrence Network	91.6%
CPM Quality Score Average for Embedding Based Network	89.4%

While daily community detection is effective for identifying topical clusters, it does not directly capture how these clusters evolve across time. For the purpose of news trend prediction, it is crucial to connect communities across consecutive time windows in order to form reliable insights on the evolution of news.

To achieve this in the embedding based network, each community was embedded in semantic space by computing its centroid embedding, derived as the mean of sentence embeddings of its constituent articles (Reimers & Gurevych, 2019). This representation captures the overall semantic content of a community and enables comparison between communities across different time windows.

$$\mathbf{v}_c = \frac{1}{|A_c|} \sum_{a \in A_c} \mathbf{e}_a \qquad\qquad \mathrm{sim}(\mathbf{v}_i, \mathbf{v}_j) = \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{\|\mathbf{v}_i\| \, \|\mathbf{v}_j\|}$$

Equations for comparing communities

Each community c was represented by its centroid vector Vc, the mean of embeddings ea for all articles a∈Ac (left equation), and pairwise cosine similarities between centroids of consecutive days were then computed using the dot product of these vectors (right equation).

Communities were compared across consecutive days using cosine similarity between centroid embeddings. Each community on day t+1 was linked to its most similar match from day t, with only connections above an elbow-determined threshold retained to ensure continuity of genuine storylines (Rossetti & Cazabet, 2018).

In contrast to the embedding based network, the co-occurrence networks were constructed around named entities extracted from the corpus. This structure captures the relational backbone of news stories, where narratives are often driven by repeated associations among key actors and concepts (Newman, 2001).

To track continuity over time, communities were converted into bags of canonical entity keys (e.g., “John Doe” vs “President Doe”), then vectorized with GPU-accelerated TF-IDF to weight discriminative entities and down-weight common ones. Inter-community similarity was measured via cosine similarity of TF-IDF vectors, with thresholds applied as in the embedding-based networks.

To standardize and track topics across the full temporal range of the dataset, a global community ID was assigned. Communities in the first time window were initialized with unique global IDs. As time progressed, linked communities that passed the similarity threshold inherited their predecessor’s global ID, thereby forming a continuous “global community” across days. Unmatched communities were assigned new global IDs.

This method reconciled fragmented daily communities into coherent longitudinal topics, forming the basis for identifying emerging, persisting, or declining trends. By linking communities through semantic continuity, it captures temporal dynamics of collective attention rather than static snapshots (Leskovec, Backstrom & Kleinberg, 2009).

3.4 Centrality Calculation

To identify influential articles and sources in the daily news networks, we computed four standard centrality measures: degree, betweenness, closeness, and eigenvector centrality. These measures capture complementary aspects of influence and are widely used in social and information network analysis (Freeman, 1979; Wasserman & Faust, 1994).

With respect to the scope of the dissertation, each centralities can be defined as:

- Degree centrality counts direct links, showing how broadly an article/entity connects within daily coverage and its likelihood of contributing to mainstream narratives (Kwak et al., 2010).
- Betweenness centrality identifies bridging nodes that connect separate clusters, often shaping emerging trends (Freeman, 1977; Borgatti, 2005).
- Closeness centrality measures how quickly an article/entity can reach others, highlighting spreadability or core narratives likely to become widely reported (Wasserman & Faust, 1994).

- Eigenvector centrality captures influence through connections to other important nodes, useful for spotting early stories tied to well-connected narratives.

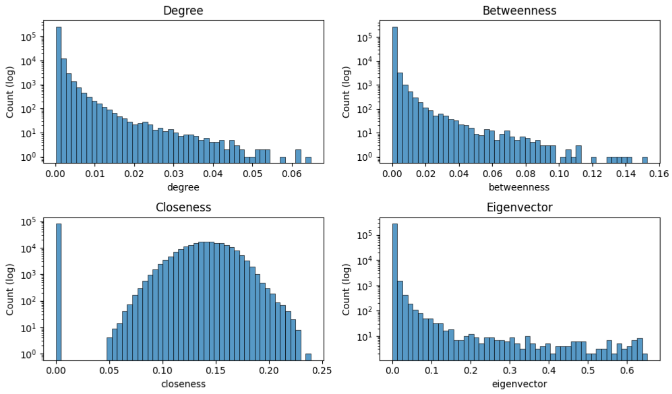

Figure 6 : Centrality Distribution for Co-Occurrence Network across all time windows.

Upon observation of both the plots in *Figure B.5* and *Figure 6* the following insights can be said:

- Degree: Most articles have very few direct connections, with only a small number acting as hubs. This suggests that only a handful of stories dominate daily coverage.
- Betweenness: A small set of articles bridge different clusters, highlighting potential "trend-setters" that connect otherwise separate narratives.
- Closeness shows that while many articles/entity remain on the network's periphery, a smaller subset is positioned at the structural core, allowing them to quickly reach diverse communities. These centrally placed articles are likely to capture and spread dominant narratives early, making them strong indicators of potential news trends.
- Eigenvector: Influence is highly skewed, with only a few articles/entity connected to other influential ones. These may represent early signals of major emerging stories.

Therefore, the network is highly unequal, where there is a few articles driving most of the structure and this is exactly what is expected of features that resemble real world scenario and these outliers are crucial signals for predicting which stories are likely to dominate or trend.

3.5 Topic Association for Global Community

3.5.1 For Embedding-Based Network

To assign interpretable topics to global communities in the network, keyword extraction using KeyBERT was used(Grootendorst, 2020) with a Sentence-BERT encoder (Reimers & Gurevych, 2019). This step is critical in news-trend prediction, as it transforms large volumes of unstructured text into concise, human-readable topic labels, enabling both interpretability and temporal tracking of emerging narratives (Blei et al., 2003; Grootendorst, 2020).

The encoder all-MiniLM-L6-v2 was chosen because it captures the meaning of text while being fast and efficient (Reimers & Gurevych, 2019). Unlike simple word counts, it understands context, so keywords reflect real meaning.

For each community, all of texts were combined into one large document so that the extracted keywords would reflect the overall discussion of the group rather than just individual pieces of text (Newman, 2006). To find possible keyword candidates, I used a CountVectorizer that looks at both single words (unigrams) and two-word phrases (bigrams), giving a good balance between informativeness and computational efficiency.

Keywords were then selected with Maximal Marginal Relevance (MMR) (Carbonell & Goldstein, 1998), balancing relevance (representativeness of the community) with diversity (avoiding redundancy). In other words, picking keywords by balancing how relevant they are to the main topic and how different they are form each other. This is particularly important in news, where communities often contain multiple storylines (e.g., political, economic, and social dimensions of the same event). These findings were then added as the global_community_title attribute.

3.5.2 For Co-Occurrence-Based Network

The extraction of keywords combined semantic embedding–based selection with frequency-based ranking to balance representativeness and salience. Using the SentenceTransformer model (all-mpnet-base-v2; Reimers & Gurevych, 2019), entity phrases were embedded into dense vector spaces, and the Maximum Marginal Relevance (MMR) algorithm was applied through the top5_by_embeddings_mmr function. MMR prioritizes phrases that are both central to the community's semantic centroid and diverse in content, ensuring that multiple facets of the discourse are represented. Complementing this, the top5_by_frequency function identified the most frequently occurring entities within each community. Frequency is an important signal in news data, as repeated mentions often correspond to prominence and public salience (Lazer et al., 2021). Together, these two approaches ensured that extracted keywords were not only semantically representative but also aligned with the most visible terms in global discourse.

To improve interpretability and reduce redundancy, subsumed phrases were removed (e.g., retaining 'John Doe' instead of both 'John' and 'John Doe'), and results were stored as global_community_title. Combining semantic centrality, lexical prominence, and redundancy filtering produced concise community-level labels, bridging structural clustering with thematic annotation to yield robust topic representations for tracking and forecasting news trends.

3.6 Exploratory Data Analysis and Feature Engineering

Following the construction of daily news co-occurrence and embedding based networks, then the computation of centrality measures, the next step is to interpret information that we have extracted to reveal the structural roles of topics in shaping information flow. Since the overarching goal of this project is to forecast news trends, the analysis is framed not only around describing which topics appear central on any given day, but also around why their structural position is meaningful for predicting diffusion, persistence, and eventual dominance in the news cycle. By examining the top-ranking clusters for each metric, we can disentangle transient noise from structural signals and thereby build hypotheses about the short-term and medium-term trajectories of news narratives.

3.6.1 Understanding Centralities

Centrality Type	Co-Occurrence Top Cluster	Embedding Top Cluster
Degree	“['washington coca cola', 'jhoan duran east', 'alexander steele', 'predator', 'rafales', 'donald trump', 'white house', 'america', 'fed']"	'cpi inflation, khari foods, setup ecoflow, test cnn, announces settlement'
Betweenness	“['washington coca cola', 'jhoan duran east', 'alexander steele', 'predator', 'rafales', 'donald trump', 'white house', 'america', 'fed']"	'cpi inflation, khari foods, setup ecoflow, test cnn, announces settlement'
Closeness	“['washington coca cola', 'jhoan duran east', 'alexander steele', 'predator', 'rafales', 'donald trump', 'white house', 'america', 'fed']"	'cpi inflation, khari foods, setup ecoflow, test cnn, announces settlement'
Eigenvector	“['washington coca cola', 'jhoan duran east', 'alexander steele', 'predator', 'rafales', 'donald trump', 'white house', 'america', 'fed']"	'cpi inflation, khari foods, setup ecoflow, test cnn, announces settlement'

Table with with the global community title

For both types of networks, global community title of the highest globally weighted centralitiy for each centrality type. It was calculated by implementing a two-stage aggregation that first computes, for each day and global community title, the mean centrality and item count, then aggregates across days by summing centrality and averaging daily counts of the global community title, derives a weighted score (weighted_d = centraltiy_value × count), and finally ranks titles by cumulative centrality and average volume.

In the Co-Occurrence network, all centrality measures (degree, betweenness, closeness, eigenvector) highlighted the same dominant cluster: Washington Coca Cola, Jhoan Durán East, Alexander Steele, Predator, Rafales, Donald Trump, White House, America, and the Fed. This shows that a small set of entities dominated visibility, connectivity, and systemic importance. Political anchors (Trump, White House, Fed) connected broad narratives; corporate actors (Coca Cola) linked business and regulation; defense terms (Rafales, Predator) reflected geopolitical themes; and lesser-known figures (Durán East, Steele) gained centrality when tied to larger conversations. The consistency across measures indicates a highly concentrated, centralised news environment.

In the Embedding-based network, the same keywords dominated across all measures: CPI inflation, Kharj foods, setup Ecoflow, test CNN, and announces settlement. CPI inflation reflected macroeconomic drivers; Kharj foods and Ecoflow highlighted business and innovation; and "announces settlement" indicated litigation and regulatory activity. "Test CNN" was an outlier, traced to a watermark/SEO artefact. Overall, the network consistently captured economic, corporate, and regulatory narratives as structurally dominant.

3.6.2 Visualising the News in a Graph

The goal is to visualise news articles/entities in a 2D scatter plot to track how topical structures evolve over time. Each point represents an article or entity, with proximity reflecting semantic similarity. By colouring points by global community and segmenting by day, it becomes possible to see when topics emerge, merge, or fade. To maintain readability, only the top five communities that persist across the most days are shown, balancing clarity with temporal dynamics.

Articles/entities were first mapped into high-dimensional vectors using the transformer model all-mpnet-base-v2, which captures semantic meaning beyond exact wording. This ensures that articles describing the same event remain close in vector space, enabling coherent clustering and clearer visualisation of evolving narratives. Two dimensionality reduction methods to create 2D projections were used: **UMAP and t-SNE**

Method	What & How	Positives	Negatives
UMAP	Builds a graph of how points connect, then lays it out in 2D while keeping both small and big patters.	• Much faster, works well on large data • Keeps local + global structure • Good default settings	• Small clusters sometimes less crisp. • Results can vary with randomness
t-SNE	Finds neighbours for	• Excellent at	• Slow on big datasets

	each point and arranges them so close points stay close in 2D.	showing tight clusters. • Very clear separation of small groups	• Distorts global structures • Needs careful tuning

Table with difference between UMAP and t-SNE.

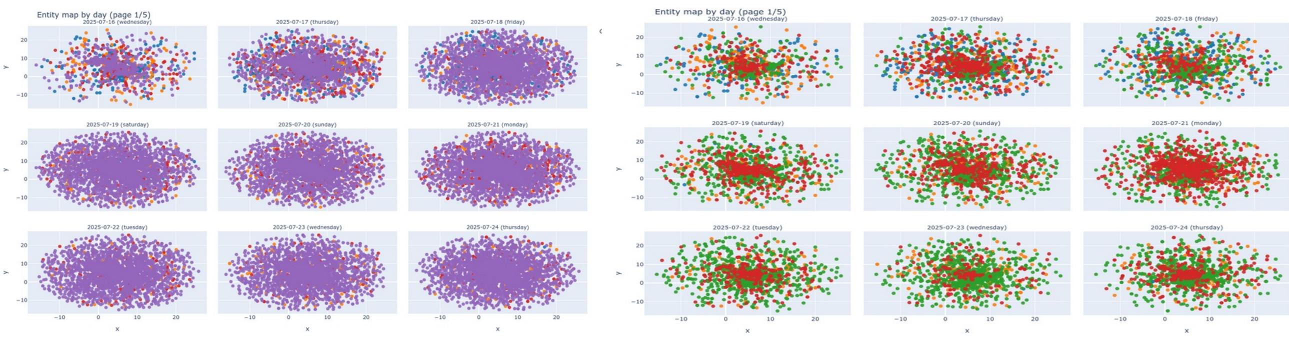

Figure 7a & 7b : UMAP Co-Occurrence entity scatterplot 7 days (right filters purple out)

Legend Information:

Blue	Alvotech nasdaq, david friedland...
Orange	Megadalena frech, sudan guterres...
Green	Medicinova nasdaq, jakks pacific...
Red	Samsung electronics, alternativ...
Purple	Washington coca cola, jhoan duran...

When plotting the entities form co-occurrence across days, the UMAP projections show that the purple community dominates every time slice, consistently spreading across most of the embedding space and reflecting a sustained focus on a single narrative. When this community is filtered out, other clusters such as red and green remain clearly visible and persist across multiple days, indicating secondary narratives that are stable over time, while smaller groups (e.g., blue and orange) appear sporadically, pointing to short-lived or less influential topics. From a trend prediction perspective, the growth and persistence of the red and green clusters highlight them as early indicators of emerging narratives, while the sporadic groups suggest fading relevance.

However, a key limitation of UMAP is evident: because the algorithm centres the embedding space each time, the absolute position of clusters across days is not directly comparable. As a result, the t-SNE dimensionality reduction to 2D was also applied for better visualization purposes.

Figure 8a & 8b : t-SNE Co-Occurrence entity scatterplot 7 days (right filters purple out)

Legend Information:

Blue	Alvotech nasdaq, david friedland...
Orange	Megadalena frech, sudan guterres...
Green	Medicinova nasdaq, jakks pacific...
Red	Samsung electronics, alternativ...
Purple	Washington coca cola, jhoan duran...

t-SNE projections show the purple community dominating the embedding space and masking smaller groups, though local structures persist: the green cluster appears compact on the right, the red cluster remains dense and central, and the blue and orange groups are smaller and peripheral. Filtering out purple reveals red and green as stable narratives, orange as a brief spike on 16–17 July, and blue as weak and scattered. Unlike UMAP, t-SNE highlights tight local clusters, making the persistence of red and green clear; for trend prediction, these represent long-term drivers, while orange reflects a short-lived surge.

The UMAP projection of **article embeddings** (*Figure B.6 along with different legends*) shows distinct clusters that remain relatively stable across days. The red cluster anchors the lower space as a persistent narrative, the green remains stable at the top, and the purple is compact and isolated to the right, each reflecting long-running storylines. The orange cluster varies more, peaking early (July 16–18) before dispersing, suggesting a short-lived event-driven trend, while the small blue cluster contributes little overall. By July 22–24, orange and blue begin overlapping with red and green, indicating emerging cross-narratives that may evolve into broader trends. Which is a key signal for prediction. A t-SNE projection was also applied for comparison.

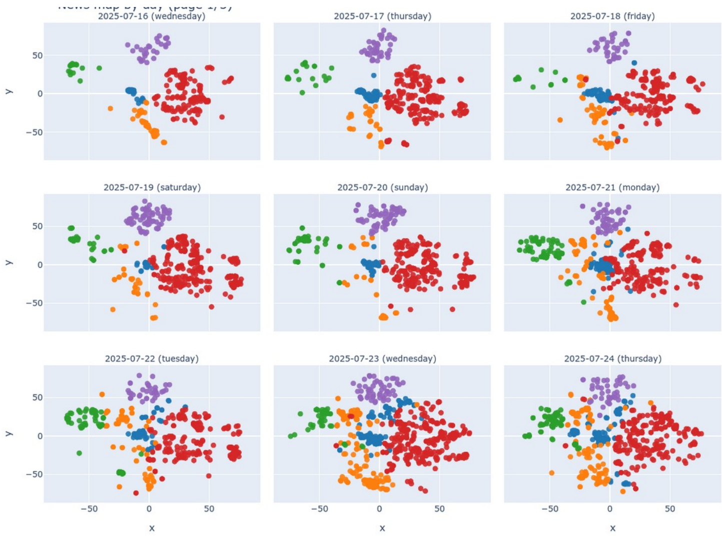

Figure 9 : t-SNE Article Embedding Scatterplot (7 days)

Legend Information:

Blue	Congress, crypto, btc surges, buyout…
Orange	Cpl inflation, khari foods, setup ecoflow….
Green	Gaza, food nutrition, strikes demand...
Red	Shares nokia, holdings channel reports…
Purple	Transmission porsche, panels missing...

The t-SNE projection (*Figure 9*) shows well-separated, compact clusters that evolve across days, reflecting both stable and shifting narratives. The red cluster is the largest and most persistent, dominating the centre throughout, while the purple cluster remains dense above the centre and the green cluster stable on the left with minor density changes. The orange cluster is dispersed and short-lived, peaking around 16–18 July 2025 before fragmenting, and the small blue cluster gradually integrates with others by 22–24 July 2025. Later, overlaps between orange, blue, and red suggest emerging blended storylines.

Compared with UMAP, **t-SNE emphasizes the tightness of local groups**, making the persistence of dominant narratives (red, purple) and the gradual convergence of smaller ones (blue, orange) more apparent, which is valuable for identifying both entrenched and emerging news trends.

3.6.3 Visualising News Flow

After computing degree, betweenness, closeness, and eigenvector centrality for each community, these measures were combined into a single feature for predictive modelling. Since the metrics overlap, using all four separately risks higher dimensionality and multicollinearity, which can weaken model generalisation (Jolliffe & Cadima, 2016).

To address scale differences, centrality measures were min–max normalised per community per day, then combined using PCA. The first component (PC1), denoted y, captured the dominant shared variance and was used as a composite centrality score indicating structural importance.

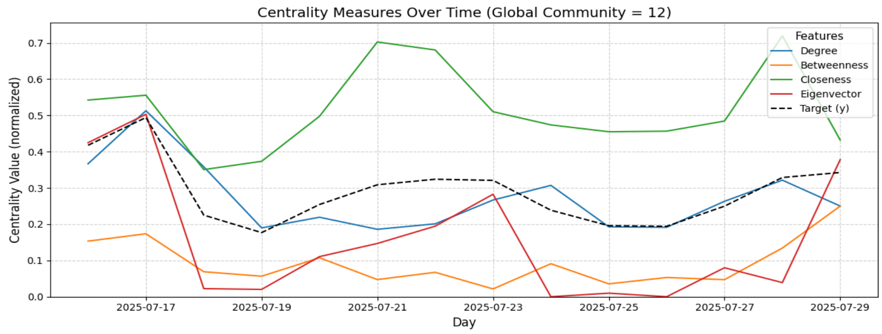

Figure 10 : Centrality Measures over Time

Figures B.7 and *10* show the temporal evolution of centrality measures against the composite feature y for a selected community. While individual scores fluctuate, y smooths these into a stable trajectory, highlighting structural importance and avoiding distortions from short-lived spikes (e.g., sudden betweenness rises). PC1 explained 68.15% of the variance (average across co-occurrence and embedding networks), confirming that one component effectively summarises degree, closeness, betweenness, and eigenvector centrality.

The feature loadings of PC1 (as per Figure 11 and Figure 12) show that y is most strongly influenced by closeness then followed by degree and eigenvector centralities, while betweenness contributes to a lesser extent for both cases of the network. This suggests that communities important for news trend dynamics tend to be those that are well-connected, accessible within the network, and embedded in influential clusters, rather than those that merely act as bridges.

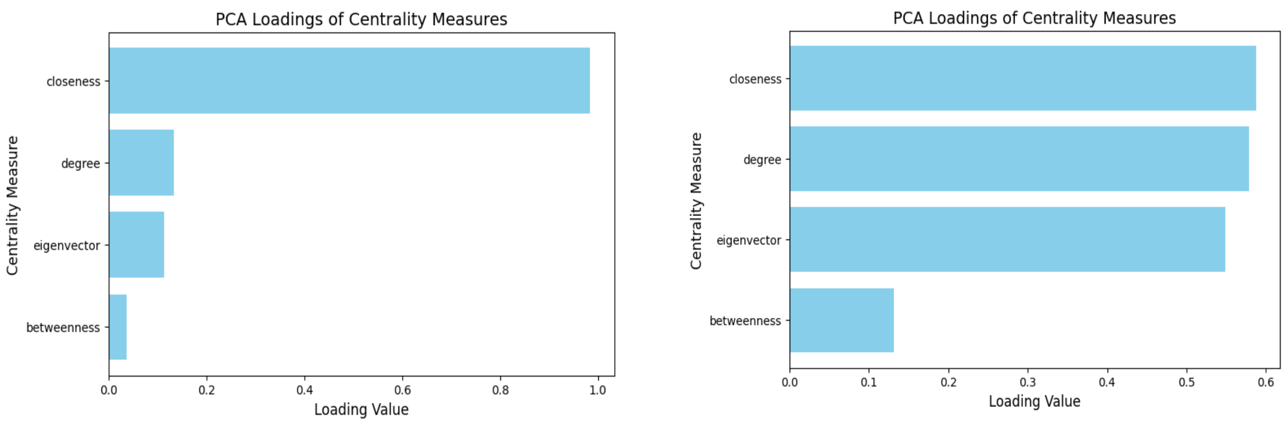

Figure 11 & 12 : PCA loadings of centrality measures in co-occurrence and embedding networks

In the context of news trend forecasting, the creation of the y column thus enables the predictive model to incorporate network-based importance in a condensed yet comprehensive way. By reducing dimensionality and noise while retaining explanatory power, this feature construction step ensures that the subsequent predictive analysis is both efficient and interpretable. A “ycombined” is made which is the median by doing a group by operation with the day and global community.

3.7 Forecasting & Modelling

3.7.1 Method 1: Predicting Trending News Communities

Method 1 forecasts trending news communities using network-based signals of information diffusion. Stories gain momentum through their position in a co-occurrence network, so features such as closeness (reachability), betweenness (bridging role), and eigenvector centrality (connection to influential stories) capture early signs of virality. These features, along with lagged values, are fed into classifiers such as Logistic Regression, Random Forests, and boosting models to predict whether a topic enters the next day’s top-trending set. This approach formalises the idea that stories trend not just through popularity, but through structural placement enabling rapid spread (Freeman, 1977; Newman, 2018; Lazer et al., 2021).

3.7.1.1 Results for Co-Occurrence Network

A community was labelled ‘trending’ if it entered the Top-10 by closeness centrality the following day, with closeness reflecting how quickly information can spread from a node to the rest of the network (Newman, 2018). The training dataset was built from daily aggregates of degree, betweenness, closeness, eigenvector centrality, and entity diversity, with lag-1 features included to capture short-term persistence by appending each community’s $t-1$ attributes to its t row. A stratified train/test split was used to preserve the ~5% positive cases across both sets, ensuring reliable evaluation despite class imbalance.

All six models achieved strong discrimination, with AUROC scores between 0.92 and 0.96 and AUPRC scores between 0.69 and 0.81. AUROC is the ability to distinguish positives from negatives overall and AUPRC is the ability to detect rare true positives with few false alarms.

Note: Boosted Trees here include XGBoost, LightBGM, Catboost for the above table.

Model Type	AUROC	AUPRC	Observation
Random Forest	0.964	0.812	Attained the best in terms of scores
SVM	0.920	0.8001	Second best when compared with the rest of the models
Logistic Regression	0.955	0.694	Slightly underperformed on AUPRC but

			still delivered excellent recall
Boosted Trees	0.93 – 0.95	0.74 – 0.77	Achieved balanced AUROC and AUPRC

Table : AUROC and AUPRC summary

Model Type	TrueNegative	FalsePositive	FalseNegative	TruePositive
Logistic Regression	454	14	3	21
Random Forest	463	5	3	21
XGBoost	461	7	3	21
CatBoost	461	7	3	21
LightGBM	460	8	3	21
SVM	463	5	3	21

Table : Confusion matrix of all models used.

All models are very good at detecting trending news (high recall, very few missed cases). Random Forest and SVM give the cleanest and most reliable signals, while Logistic Regression errs on the side of caution, flagging more stories as trending even if some are not (reducing trend capture misses). The boosting models (XGBoost, LightGBM, CatBoost) sit between these extremes, still strong, but slightly noisier than Random Forest or SVM.

With access to a much larger news dataset, model accuracy could be scoped more clearly. Since only a small fraction of news items actually become trending, the dataset naturally exhibits a class imbalance toward "not trending." This imbalance may bias models toward predicting non-trending outcomes. Hence, balancing precision (avoiding false trending predictions) and recall (capturing all real trends) is critical for robust forecasting.

Additionally below is the top 3 news communities that was detected by Logistic Regression to be trending:

- ['washington coca cola', 'jhoan duran east', 'alexander steele', 'predator', 'rafales', 'donald trump', 'white house', 'america', 'fed']
- ['medicinova nasdaq', 'jakks pacific', 'cooper', 'speec', 'nyax', 'securities exchange commission', 'institutional investor owned', 'sec institutional investor', 'fund owned shares', 'nyse']
- ['magdalena frech', 'sudan guterres', 'bulldogs', 'israelap', 'khan', 'palestinians', 'israeli', 'hamas', 'gaza']

Topics with these keywords have high trend setting potential for the present. It should be understood that the trends for news changes on a daily basis and the real world is very volatile. On investigation, it was noticed that the trends predicted is guaranteed for atleast 1 week when verifying with the news post the date of the dataset.

3.7.1.2 Results for Embedding-Based Network

In this variant, the underlying network was derived from semantic embeddings of textual entities rather than graph-structural co-occurrences. Communities were again aggregated on a daily basis, and the label was defined by Top-10 degree centrality on the following day. This setup complements the co-occurrence graph by capturing semantic similarity beyond surface tokens, while preserving longitudinal community continuity.

Model Type	AUROC	AUPRC
Logistic Regression	0.899	0.524
Random Forest	0.862	0.420
XGBoost	0.861	0.341
LightGBM	0.842	0.378
CatBoost	0.863	0.531
SVM	0.831	0.337

Table with summary of AUROC and AUPRC across models.

Overall, AUPRCs are moderate, with Logistic Regression and CatBoost providing the best precision–recall trade-off. The relatively lower scores likely reflect dataset limitations and the nature of news, where trending items form a small fraction. Expanding to larger, multi-year datasets would mitigate this imbalance and improve model precision and quality.

Model Type	TrueNegative	FalsePositive	FalseNegative	TruePositive
Logistic Regression	190	14	5	13
Random Forest	200	4	14	4
XGBoost	195	9	10	8
LightGBM	197	7	13	5
CatBoost	192	12	10	8
SVM	202	2	17	1

Table consisting confusion matrix details for all models

- In the case of embedding based, Logistic Regression is the most effective at catching real trending news, even at the cost of some false alerts.

- Random forest and SVM are too conservative, nearly filtering out all trending content.
- The boosting models (XGBoost, LightGBM, CatBoost) sit in the middle, better than Random Forest/SVM at catching trends, but still limited in recall compared to Logistic Regression.

This highlights a core challenge: since only a small portion of news truly becomes trending, models are naturally biased toward predicting “not trending.” **For a practical system, recall (not missing trends) may be more valuable** than extreme conservatism, as missing a major trending story could be costlier than raising a few extra alerts.

The top 3 news communities that was detected by Logistic Regression to be trending:

- transmission porsche, specification inline, installing bilstein, running cost, panels missing.
- ryder cup, open harris, logins paywalls, bash taps, increase durability.
- gaza deaths, food nutrition, strikes demand, treating weak, children enablers.

These keywords signal trend potential, but news shifts daily and is volatile.

3.7.1.3 Method 1: Conclusion

Method 1 shows that forecasting trending news communities is feasible by combining structural and semantic signals. Co-occurrence and embedding-based networks both enabled classifiers to detect early signs of virality: Random Forest and SVM gave strong precision, Logistic Regression maximised recall, and boosting models offered balanced but weaker recall. This highlights the trade-off between missing true trends (false negatives) and raising extra alerts (false positives) in fast-moving news (Freeman, 1977; Newman, 2018; Lazer et al., 2021). Class imbalance remains a challenge, as only a small fraction of stories trend, biasing models toward negatives. Larger, more diverse datasets could improve calibration, but overall, this method offers a systematic framework that considers both popularity and structural positioning as key drivers of diffusion potential.

In a production setting. One would input a collection of news from “t” present day, feed into the parallel pipeline that considers both Co-occurrence and Article embeddings which would turn it into its respective networks and then gather centrality values which is then followed by the model flagging the trending setting news communities in the form of key terms and words.

3.7.2 Method 2: Forecasting News Trends with Time Series

3.7.2.1 Linear Regression Variant (Huber)

The forecasting method identifies news communities gaining or losing traction by compressing daily centralities into a composite score (section **3.6.2**) and fitting a trend line over time. The slope's sign indicates direction, and its magnitude reflects trend strength, yielding an intuitive momentum signal that can be ranked, visualised, and integrated into downstream forecasting or alerting.

To capture temporal dynamics, a per-community Huber regression was fitted to the first 60% of days, providing clear visualisation while reducing sensitivity to spikes and anomalies compared to standard linear regression. Dates were encoded as ordinal values, and the fitted model returned slope estimates representing overall trend.

To facilitate cross-community comparison, estimated slopes are normalized to the interval [0,1] using min–max scaling, producing a normalized slope (nslope). This allows communities to be ranked in terms of relative trend strength, with higher values reflecting stronger positive trajectories and lower values indicating weaker or negative growth.

3.7.2.2 Results for Co-Occurrence Network (Entities)

The top 2 news communities with the highest slopes were:

- “['washington coca cola', 'jhoan duran east', 'donald trump', 'white house', 'america', 'fed']" (slope: 0.073)
- “['alvotech nasdaq', 'david friedland', 'union pacific', 'ryt bank', 'citadel', 'marketbeat com', 'morgan stanley', 'roth capital', 'scotiabank', 'benzinga']” (slope: 0.039)

These are the keywords of interest that has high trend potential.

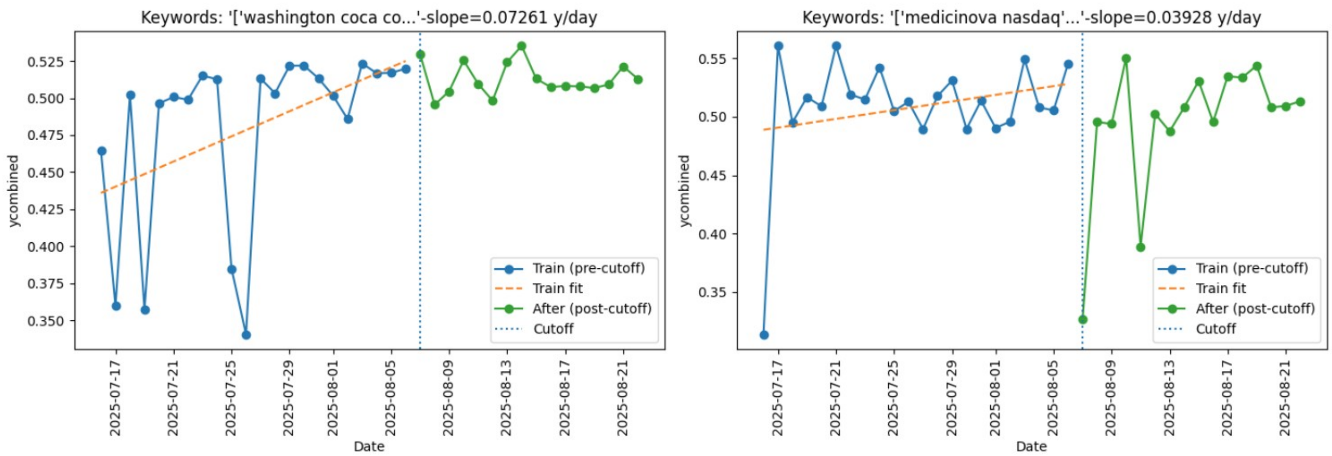

Figure 21a & 21b : Visualizing trend line with the past and future values for the top two

In Figure 21a, the fitted line shows a clear positive slope (0.07261 ycombined/day), with values stabilising around 0.5–0.53 after the cut-off, indicating sustained momentum and a strong likelihood of remaining or becoming trending—making it a priority for alerts and monitoring. By contrast, Figure 21b shows a slower rise (0.039 ycombined/day), followed by sharp dips and rebounds to ~0.5–0.55 with volatile swings, suggesting fragile but growing interest typical of event-driven coverage. Despite its volatility, it remains a valuable point of interest for actionable insights.

In conclusion for this case, one would treat topic A as s confident, continuing trend and treat Topic B as an emerging but noisy trend with a higher chance of resurfacing but also comes with the additional risk of short lived drops. For forecasting pipelines, rank by slope to surface strong movers. In the future, with a larger set of data a threshold could be devised to add more confidence to the predictions so that no unproductive news communities would surface.

3.7.2.3 Results from the Article Embedding Network

The top 2 news communities with the highest slopes were:

- 'cpi inflation, khari foods, setup ecoflow, test cnn, announces settlement' (slope: 0.066)
- 'trump ukraine, weather mission, approved tightening, target interceptor, nanoplastics' (slope: 0.0049)

These are the key terms that is predicted to have trend potential. However, upon further investigation with the below plots, the following observations were made.

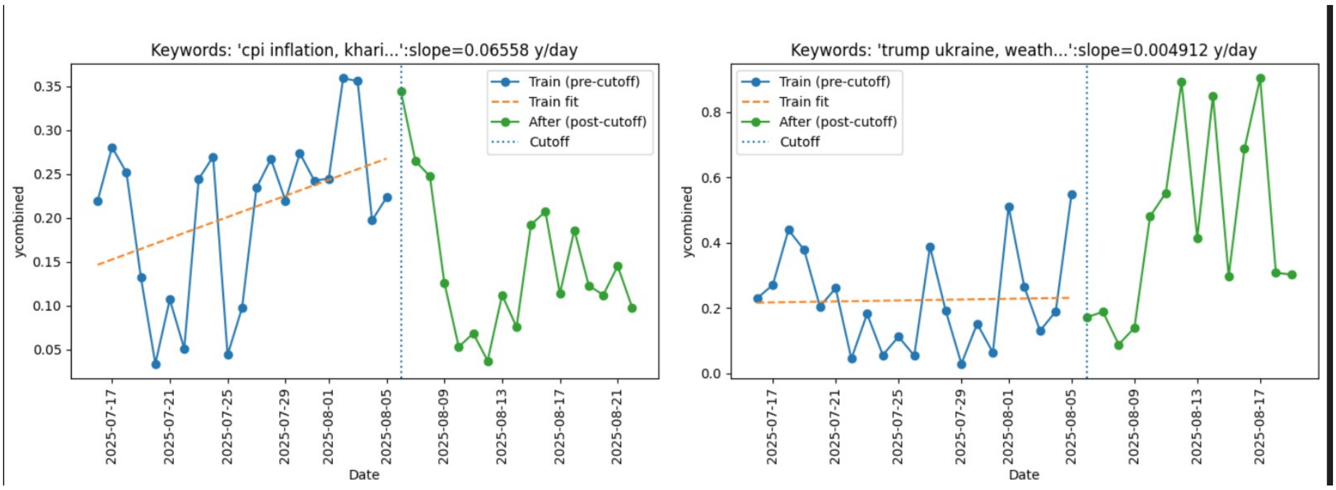

Figure 22a & 22b : Visualizing trend line with the past and future values for the top two

In Figure 22a, the training period shows a solid upward drift, but after the cut-off the level drops sharply and oscillates lower. It's a pattern typical of calendar events such as a CPI release, where intense build-up is followed by fading attention, producing a short-lived trend. In contrast, Figure

22b remains nearly flat before the cut-off but rises sharply afterwards. This is because the model fits only the first half with a single linear trend, it assigns a small slope and de-prioritises the topic just before it takes off. Similar patterns observed in other communities highlight concerns about the model's reliability.

In conclusion, using slopes from using Huber regression on the embedding communities is not suitable for forecasting news trends in our setting. It assumes a single straight line trend over the training window, so it cannot anticipate changes or post cutoff breakouts. Additionally in the real world the the time windows and the datas can change because the entire model is trained every time for additional data as we are only using the slope lines. This makes the model very unreliable and unpredictable for data of different time windows.

3.7.2.4 Improving with the Prophet Model

To handle how news really behaves, the Prophet model was adopted. It is an additive forecasting model with piecewise linear trends an automatic change point detection. In other words, Prophet can notice when the direction of a series suddenly changes, it can build in daily, weekly or yearly seasonality. It is also robust to outliers and missing days. Additionally also returns interpretable components with uncertainty bands. Making it a fitting candidate for timely and reliable trend prediction (Taylor & Letham, 2018).

To predict which news communities are about to trend for both the cases (Co-occurrence and embedding), the Prophet model is applied to each community's daily composite centrality. The trend decision here is the probability that the next step in the forecast increases. Therefore, higher the probability, the higher the trend potential.

3.7.2.5 Results for Co-Occurrence Network (Entities)

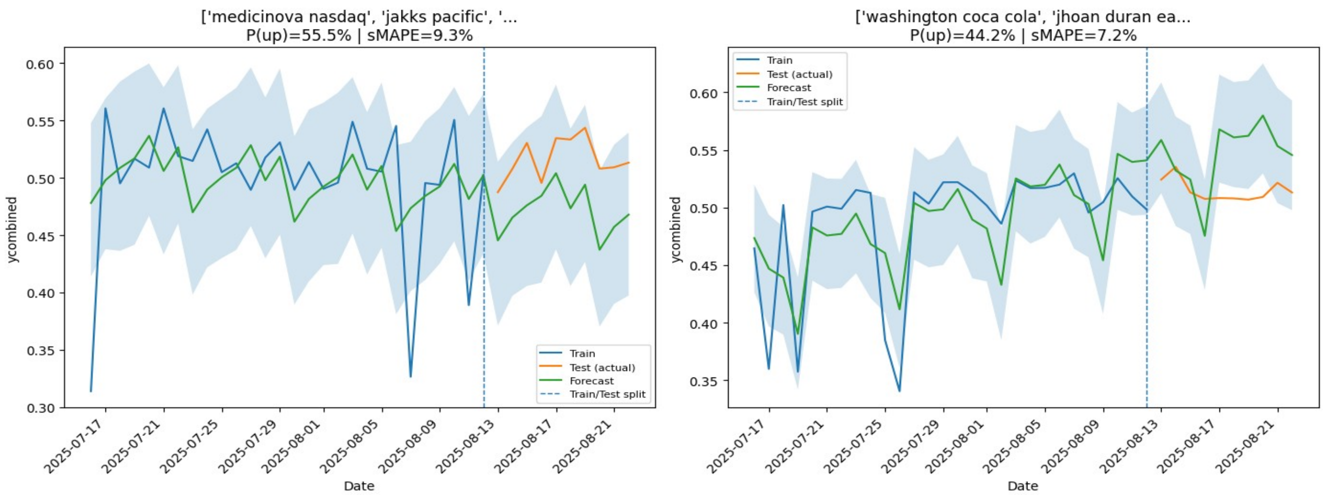

Figure 23a & 23b : Prophet predictions over original values

	Accuracy	SMAPE (lower=better)	Coverage
Figure 23a	90.67%	9.3%	0.9
Figure 23b	92.79%	7.2%	0.6
Average	91.74%	8.26%	0.75

Table : Performance of the model

Coverage here describes the share of actual test day values that fell inside Prophet’s prediction interval.

Topics of interest retrieved:

- ['medicinova nasdaq', 'jakks pacific', 'cooper', 'speec', 'nyax', 'securities exchange commission', 'institutional investor owned', 'sec institutional investor', 'fund owned shares', 'nyse'] (+55.49% probability of going up in next step)
- ['washington coca cola', 'jhoan duran east', 'alexander steele', 'predator', 'rafales', 'donald trump', 'white house', 'america', 'fed'] (+44.23% probability of going up in next step)

Figures 23a and 23b show Prophet forecasts are generally reliable but require caution with volatile series. In Figure 23a, pre-split volatility widened post-split intervals; despite a >50% rise probability, broad bands limit confidence, making it suitable for watchlisting rather than high alert. In Figure 23b, steadier variability let Prophet tighten intervals and reduce error, but with only a 44.23% rise probability and stable bands, the topic is classified as neutral to slightly downward unless new events emerge.

In conclusion, prophet is useful for news-trend prediction because it outputs a clear direction signal P(up) (Probability of next step higher than present) together with uncertainty bands, handles short and noisy daily series, and captures gradual trend shifts via changepoints (ability for the model to understand changes in trends) making “watchlist vs. hold” decisions straightforward. Its limits are that event-driven breakouts can be under-forecast, intervals may be wide when history is short, and results can vary by community and model settings.

3.7.2.6 Results from the Article Embedding Network

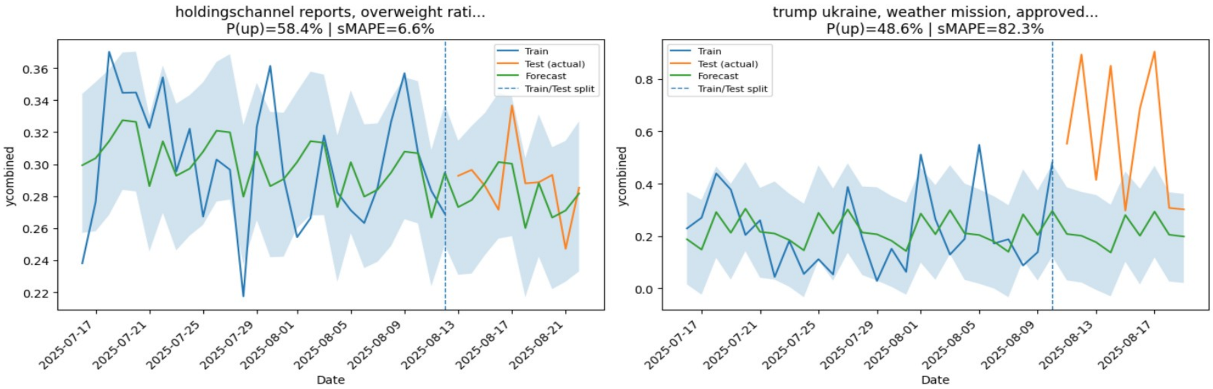

Figure 24a & 24b : Prophet predictions over original values

	Accuracy	SMAPE (lower=better)	Coverage
Figure 24a	93.36%	6.63%	1
Figure 24b	17.74%	82.25%	0.33
Average	55.55%	44.44%	0.665

Table : Performance of the model

Topics of interest retrieved:

- holdingschannel reports, overweight rating, btc semler, shares nokia, medic ballentine (+58.49% probability of going up in next step)
- trump ukraine, weather mission, approved tightening, target interceptor, nanoplastics (+48.66% probability of going up in next step)

In Figure 24a, the series remains stable around ~0.30 with minor oscillations, the forecast closely tracks this level, and all test points fall within the band (coverage=1). Errors are low, and the 58.5% rise probability suggests a gradual, persistent trend rather than a breakout, making it a candidate for incremental follow-through. In contrast, Figure 24b shows test values jumping sharply after the split while the forecast remains flat, resulting in large error. The model assigns a neutral trend (48.7% rise probability), but the reality was a major surge, likely driven by a post-split news shock absent from the training data, which the model could not anticipate.

The use of embedding based prophet model works well for steady, developing themes but is insufficient on its own for sudden, event driven spikes. A practical system should combine the Prophet model with event detectors and exogenous signals along with tighter changepoint settings which could react to faster regime shifts.

3.7.2.7 Method 2: Conclusion

Method 2 used a composite target y (PC1, 64.73% variance) from degree, betweenness, closeness, and eigenvector centralities, with closeness dominating as topic visibility. Huber-regression provided an intuitive momentum score but worked only under smooth changes, missing breakout topics. Prophet was more effective, offering piecewise-linear trends and uncertainty bands for risk-aware forecasting. On co-occurrence communities, it achieved high accuracy (≈91.7%, sMAPE ≈8.3%, Coverage ≈0.75), while on embeddings it captured steady themes but failed on sudden surges (≈17.7% accuracy). The best design is Prophet on co-occurrence networks, Huber slope as a lightweight pre-filter, and embedding forecasts enhanced with event detectors like volume or entity bursts.

4. Overall Results

The experimental evaluation demonstrated that forecasting emerging news trends is feasible using the proposed combination of network science, natural language processing and machine learning. Two methodological pipelines were examined:

4.1 Method 1 – Classification of Trending Communities

On co-occurrence networks of named entities, all classifiers performed strongly, with Random Forest and SVM giving balanced predictions and *Logistic Regression achieving the best recall (most unlikely to miss trends)*. Boosted trees (XGBoost, CatBoost, LightGBM) were competitive but slightly noisier. On embedding-based networks, results were more modest, as semantic similarity struggles with short-lived event-driven spikes; Logistic Regression and CatBoost performed best, while conservative SVM models often missed true trends. Across both network types, a precision–recall trade-off was clear: *conservative models reduce false alarms but risk missing important stories,* whereas *recall-focused models raise more false positives* but ensure key trends are detected. A preferable balance for stakeholders like journalists or policymakers, where missing a narrative can be more costly than filtering noise.

4.2 Method 2 – Time Series Forecasting of Composite Centrality

The Huber regression slope produced intuitive momentum signals on co-occurrence networks, capturing stable upward trends (e.g., US political and economic clusters) and volatile emergent ones, but underperformed on embedding-based communities where abrupt post-training surges defied linear fits. Prophet substantially improved accuracy, achieving 97.7% on co-occurrence communities by flagging sustained and gradual growth, with change-point detection and seasonality handling that better adapted to evolving cycles. On embedding networks, *Prophet worked for steady narratives* (e.g., corporate or finance clusters) but *underestimated sudden event-driven surges* (e.g., geopolitical shocks), underscoring the limits of embeddings for short-horizon forecasting. Sector-wise application of the model may offer greater stability, making this a valuable direction for future work alongside hybrid integration with event detectors or burst analysis.

4.3 Key Findings

- Co-occurrence networks were the strongest predictors, with closeness and eigenvector centrality giving stable early signals of virality. Random Forest (AUROC 0.964, AUPRC 0.812) and Logistic Regression (high recall) showed the best performance.

- Embedding-based networks added value by capturing contextual and thematic overlaps missed by co-occurrence networks, clustering events like CPI inflation or corporate settlements. Logistic Regression and CatBoost performed best (AUROC ≈ 0.89; AUPRC ≈ 0.52).
- Limitations of embeddings: they struggled with event-driven shocks, under-predicting sudden spikes (e.g., geopolitical crises), making them more reliable for gradual, theme-driven trends.
- Hybrid pipelines combining co-occurrence for stability and embeddings for context are the most reliable design for real-world forecasting.

4.4 Conclusion

This dissertation explored whether emerging news trends can be forecasted by combining network science, natural language processing, and machine learning. Using co-occurrence and embedding-based networks, community detection, and forecasting models, the study showed it is possible to anticipate which narratives are likely to gain traction.

The findings establish several core insights:

- Structural signals: Co-occurrence networks of named entities were most effective for identifying emerging communities, with high closeness and eigenvector centrality reliably indicating virality (Newman, 2018; Freeman, 1979).
- Semantic signals: Embedding-based networks added contextual depth, capturing themes like macroeconomics, litigation, and policy, but underperformed on sudden event-driven surges.
- Model trade-offs: Logistic Regression gave the best recall, Random Forest the cleanest precision, boosting models a middle ground, and Prophet outperformed linear slopes by adapting to gradual shifts with interpretable uncertainty bands (Taylor & Letham, 2018).
- Hybrid pipelines: Combining entity-based robustness with embedding-based nuance offers the most reliable balance of stability and sensitivity, reducing the risk of missing subtle but important trends.

These findings have practical value: journalists can prioritise high-impact stories before they peak, businesses can use trend signals for market intelligence and risk assessment, and policymakers can act on early-warning signals for crises or reforms (Nicholls & Bright, 2018; Lazer et al., 2021). However, limitations include the short five-week, English-dominated dataset and the difficulty of forecasting sudden event-driven surges, especially with embeddings. Future work should expand to multi-year, multilingual data, integrate external signals like social media or markets, and explore graph neural networks (GNNs) for richer temporal modelling (Xu et al., 2023).

In conclusion, this work shows that news trend forecasting is both feasible and impactful. By combining structural and semantic perspectives, it provides a framework that explains how trends emerge while giving stakeholders foresight in fast-moving information ecosystems. In an era where information outpaces attention, such predictive analytics can shift news from reactive reporting to proactive decision support.

5. Practical Implications and Competitive Advantage

This dissertation demonstrates practical value by transforming vast, fast-moving news streams into actionable foresight, offering users a strategic advantage. Journalists can anticipate and cover stories before they peak, businesses can forecast consumer sentiment and market shifts, and policymakers can detect early signals of unrest or misinformation to support proactive decision-making. The contribution lies in addressing three core challenges: filtering millions of articles to reduce information overload, forecasting tomorrow's narratives to tackle time-criticality, and reducing decision uncertainty through probabilistic, interpretable models. By shifting news analysis from reactive description to predictive insight, the framework enables earlier intervention, more efficient resource allocation, and sustained strategic advantage.

6. Future Works

Future research should use multi-year, multilingual datasets to capture seasonal cycles and global diversity, while integrating signals from social media, markets, or policy reports to strengthen predictive power. Burst detection and graph neural networks (GNNs) could better handle event-driven surges and reduce manual feature engineering, while combining Prophet with event detectors or ensembles and exploring transformer-based time-series models may enrich forecasting. Additional opportunities include improving community title detection, tracking temporal keyword changes, analysing trend decay windows (e.g., 3–14 days), and optimising data window sizes. For deployment, emphasis should be on real-time monitoring and human-in-the-loop validation to ensure forecasts remain actionable and trusted.

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Appendices

Appendix A : Code Repository

The project is accessible through the GitLab repository link below:

https://git.cs.bham.ac.uk/projects-2024-25/axx410

These are the main folders of the repository are in the "Branch_Main" folder:

- Preprocessing of the dataset is in the "*Preprocessor*" folder
- Network Creation, analysis are in the folder "*Temporal Network Analysis*"
- Forecasting related code is in the "*Temporal Network Analysis/Forecasting*" folder.
- Plots for UMAP and t-SNE is found at:

  "*Temporal Network Analysis/Generated/UltimateDataset/{NER or UMAP}/{UMAP or TSNE}*"

The code must be run on python 3 environment with a CUDA enabled GPU for a stable runtime.

Appendix B : Supporting Figures

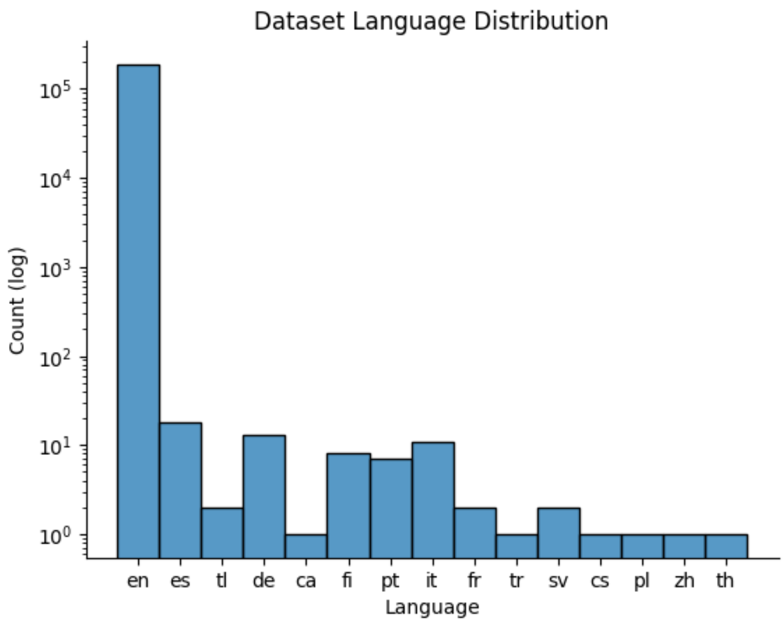

Figure B.1 : Dataset Language Distribution

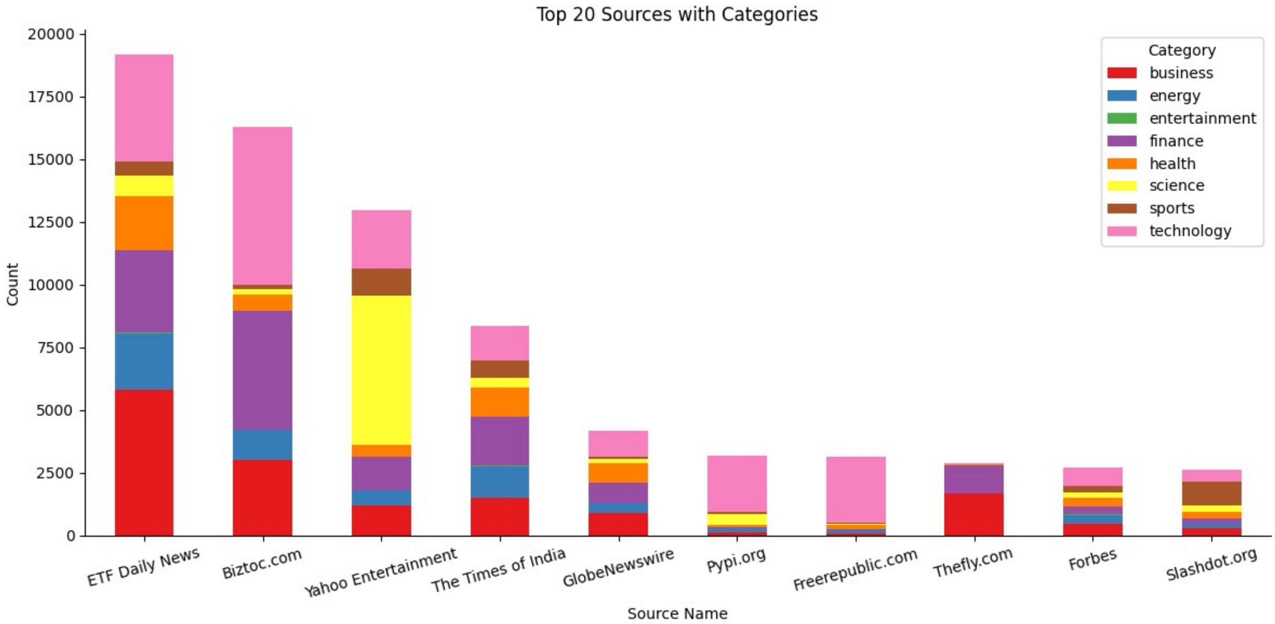

Figure B.2 : Topic(category) Coverage Per Source

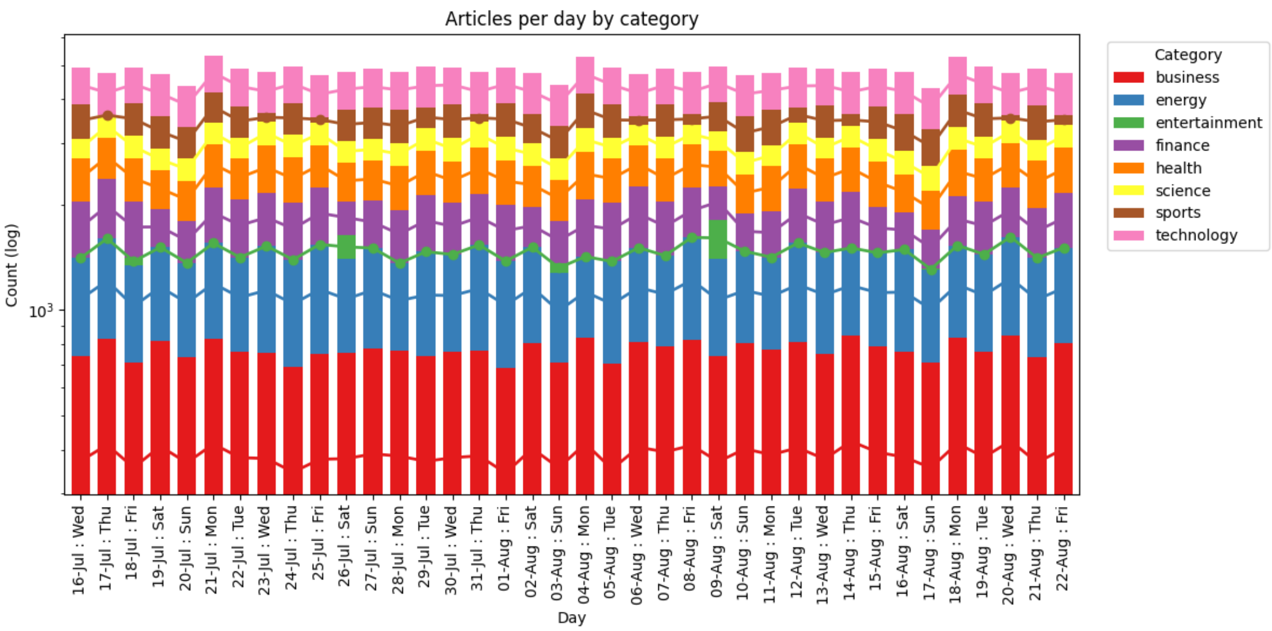

Figure B.3 : Topic(category) Coverage Per Day

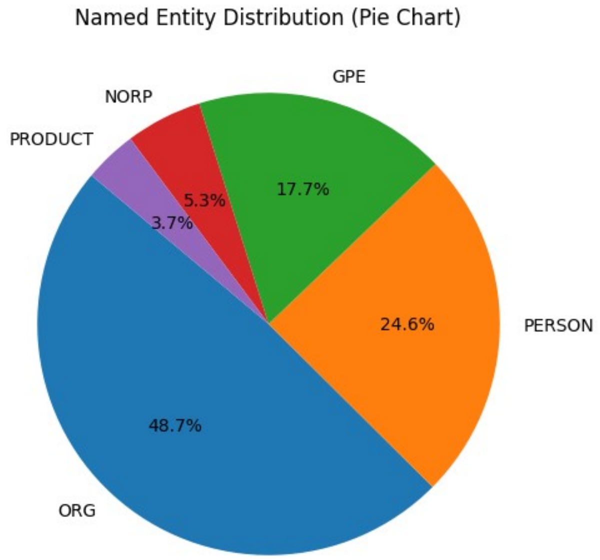

Figure B.4 : Entity Label Coverage Chart

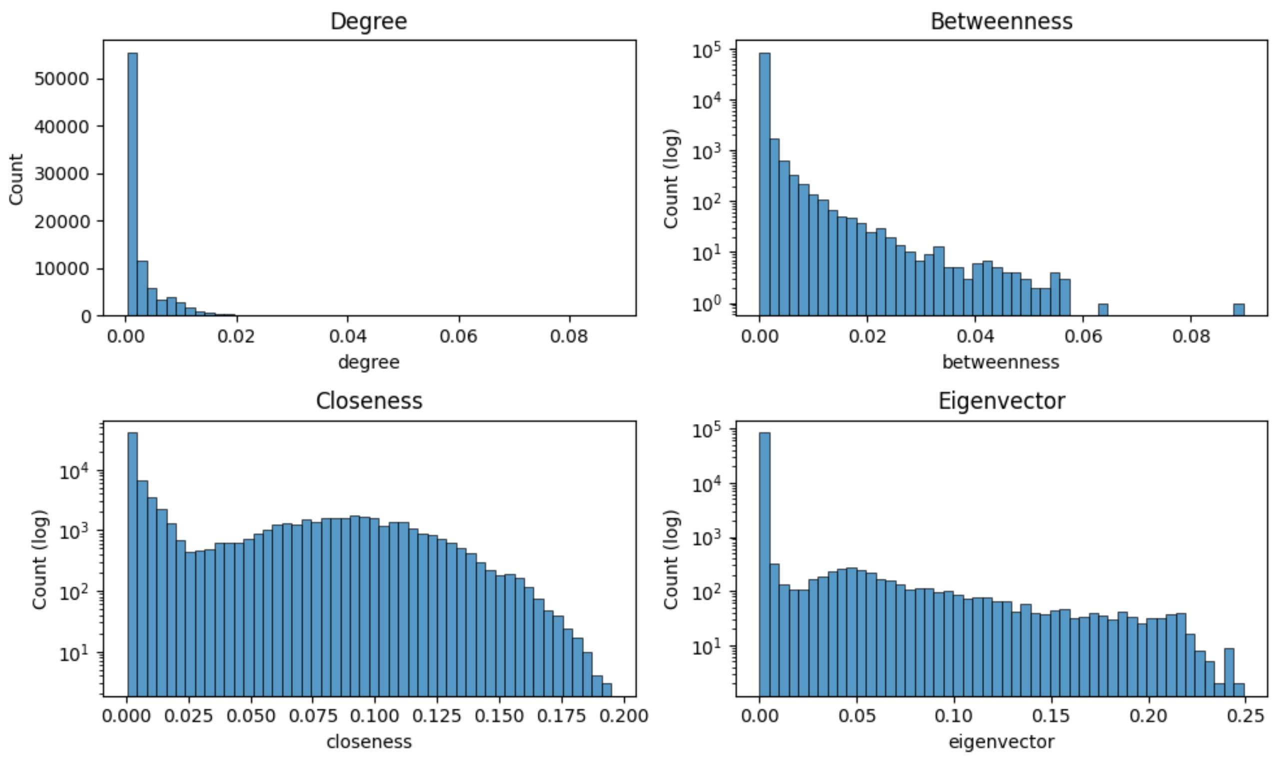

Figure B.5 : Centrality Distribution (Embedding Network)

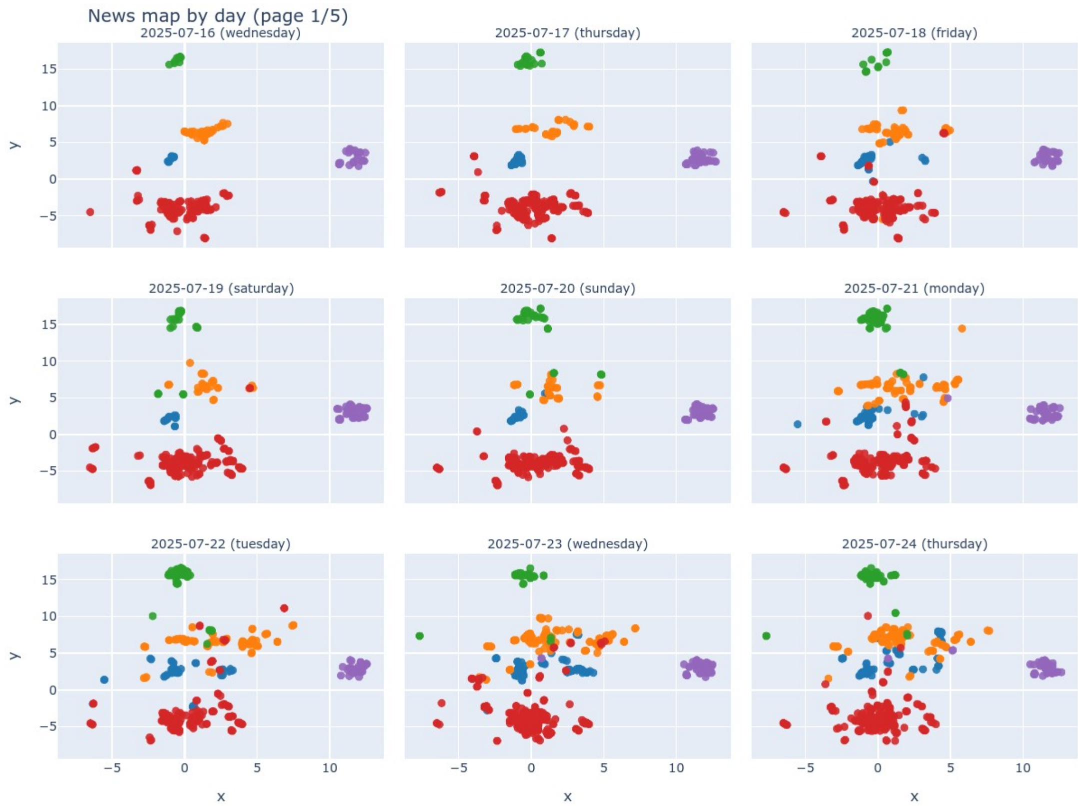

Figure B.6 : Article Embeddings UMAP Scatteplot

Legend Information for Figure B.6:

Blue	Congress, crypto, btc surges, buyout…
Orange	Cpl inflation, khari foods, setup ecoflow…
Green	Gaza, food nutrition, strikes demand...
Red	Shares nokia, holdings channel reports…
Purple	Transmission porsche, panels missing...

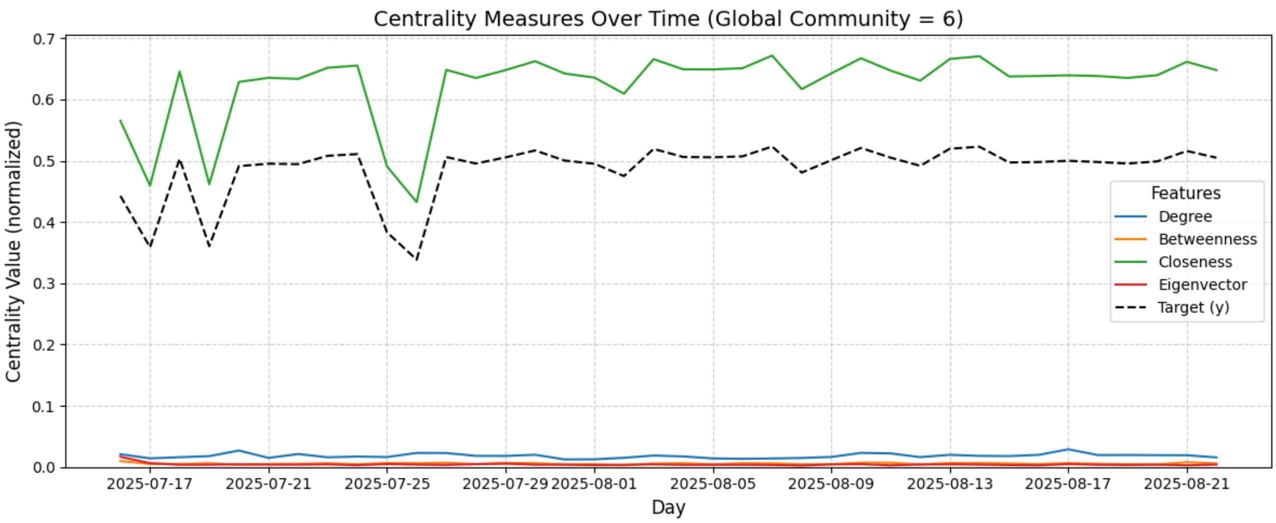

Figure B.7 : Centrality Measures over time (Co-occurrence Network)

Appendix C : Additional Figures

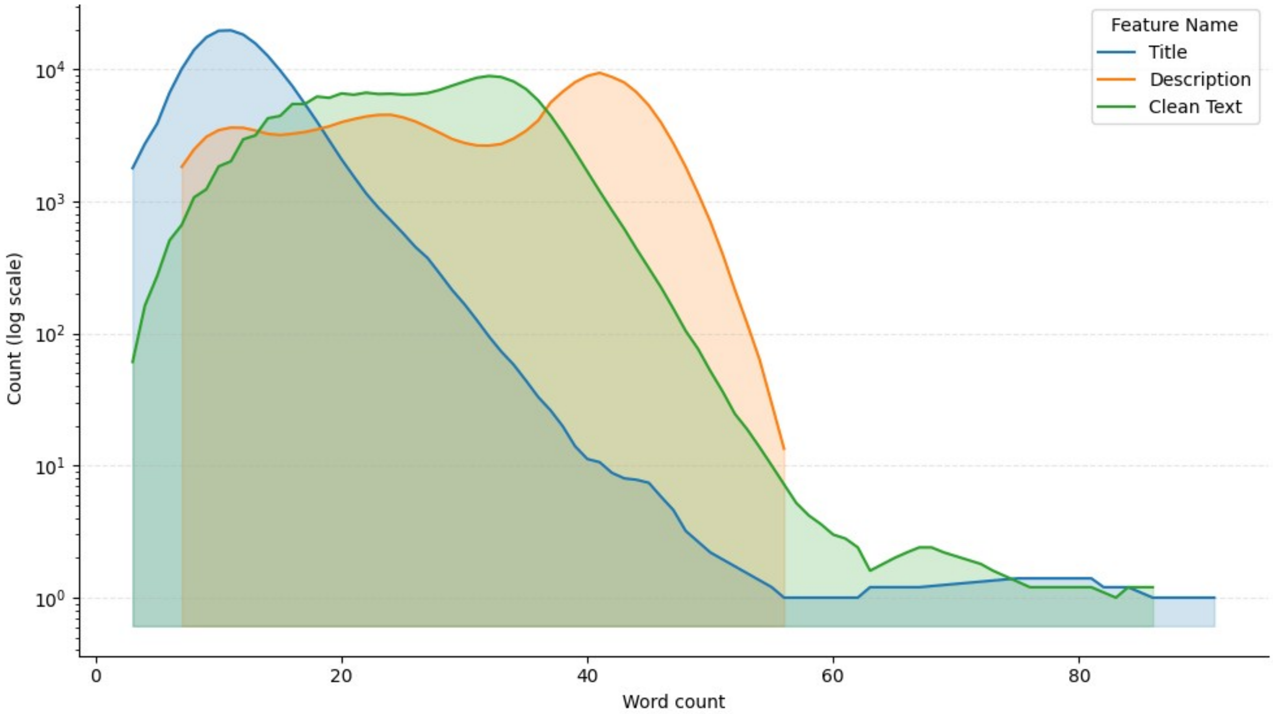

Figure C.1 : Rolling average of count of word counts for Title, Description, Clean_Text Columns

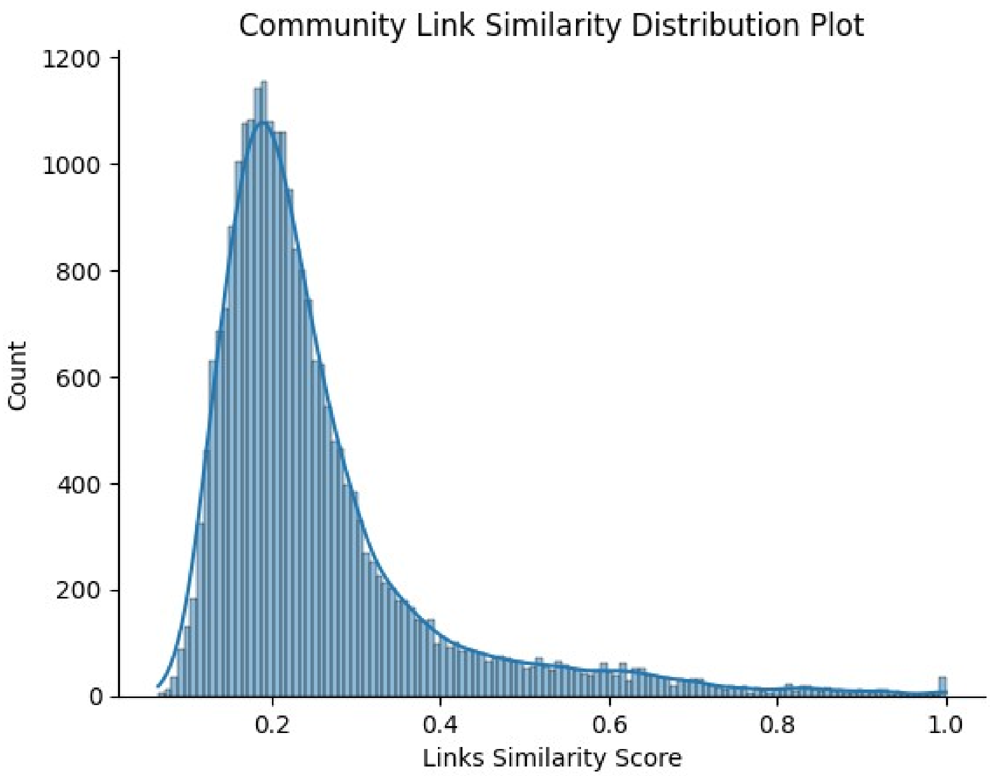

Figure C.2 : Community Linking Similarity Distribution Plot (Co-Occurrence Network)

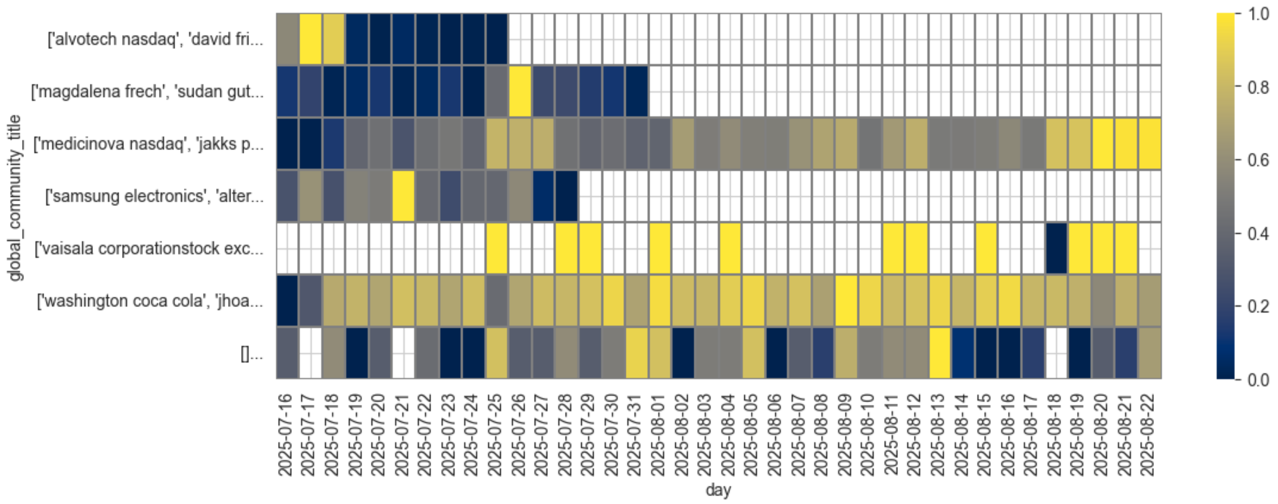

Figure C.3 : Top topic (global community title) persistence for Co-Occurrence Network

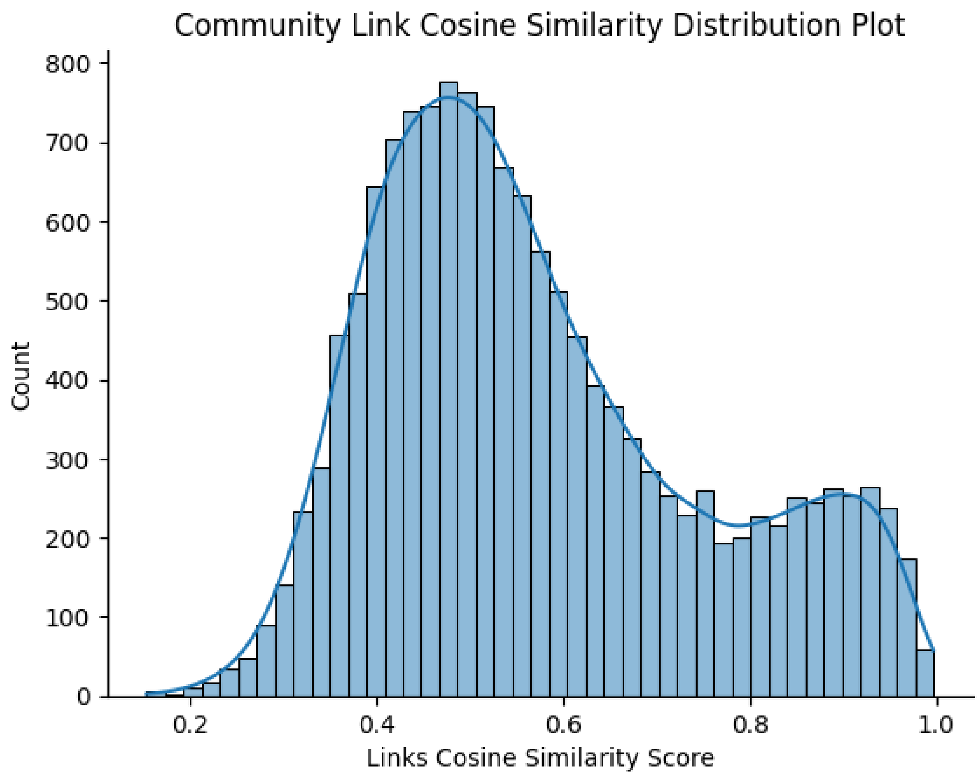

Figure C.4 : Community Linking Similarity Distribution Plot (Embedding Network)

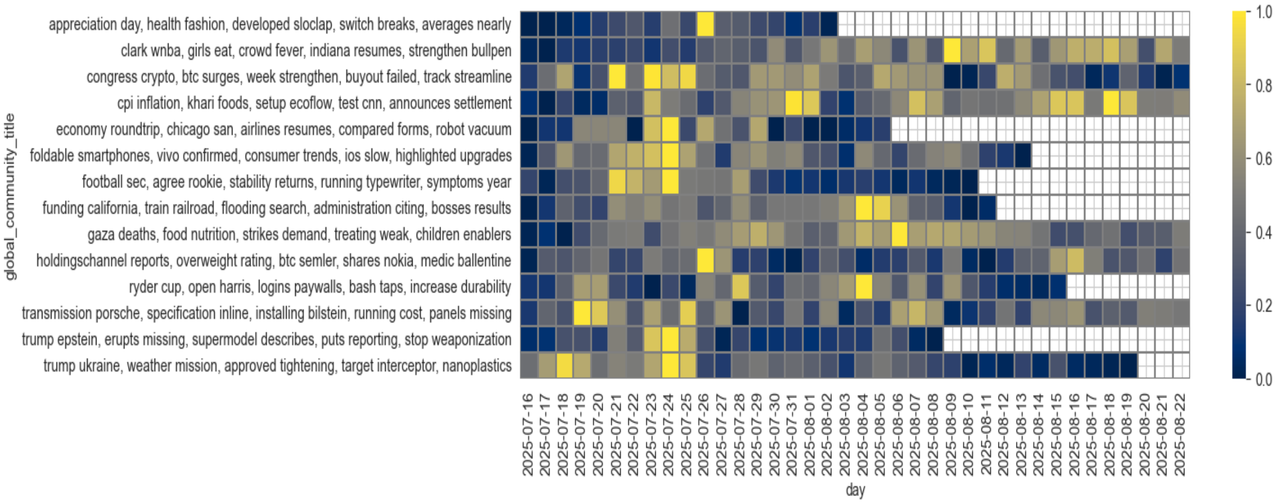

Figure C.5 : *Top topic (global community title) persistence for Embedding Network*

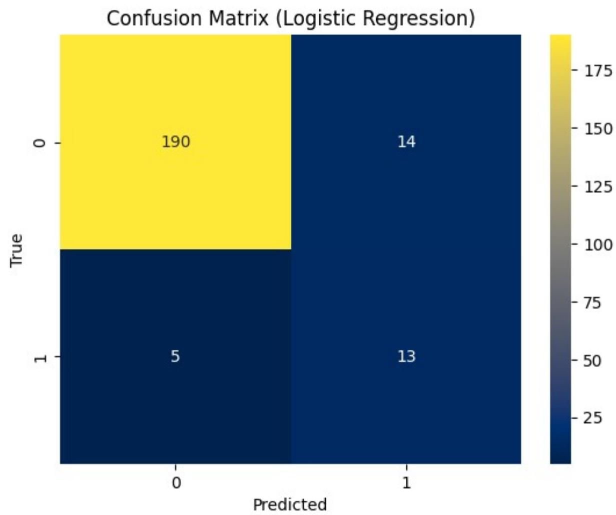

Figure C.5 : Method 1 confusion matrix for logistic regression (Embedding Network)

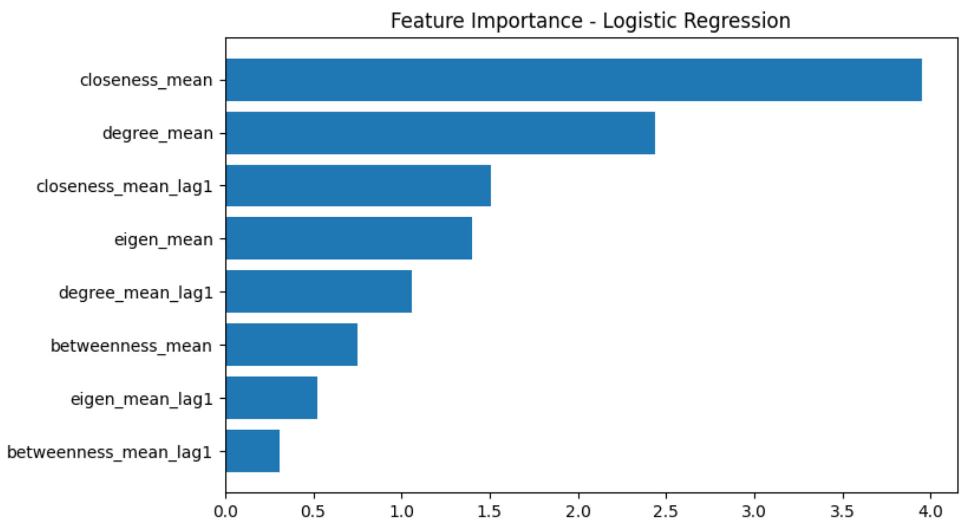

Figure C.5 : Method 1 feature importance for logistic regression (Embedding Network)

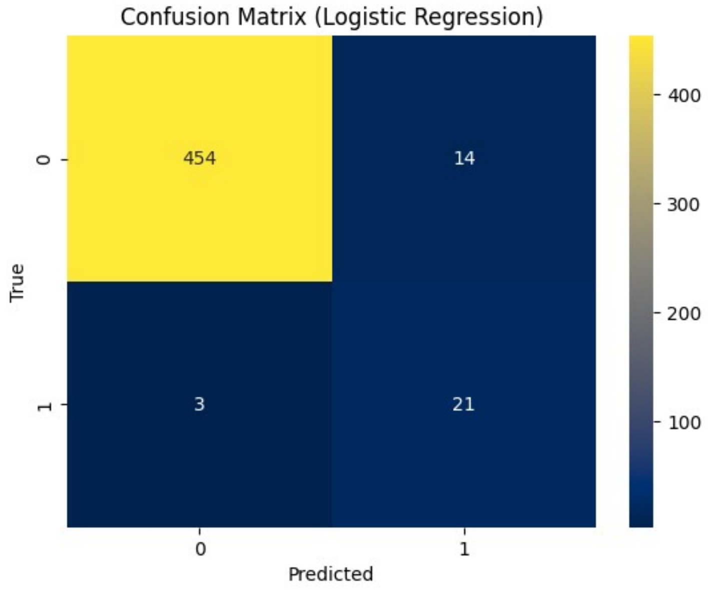

Figure C.6 : Method 1 confusion matrix for logistic regression (Co-Occurrence Network)

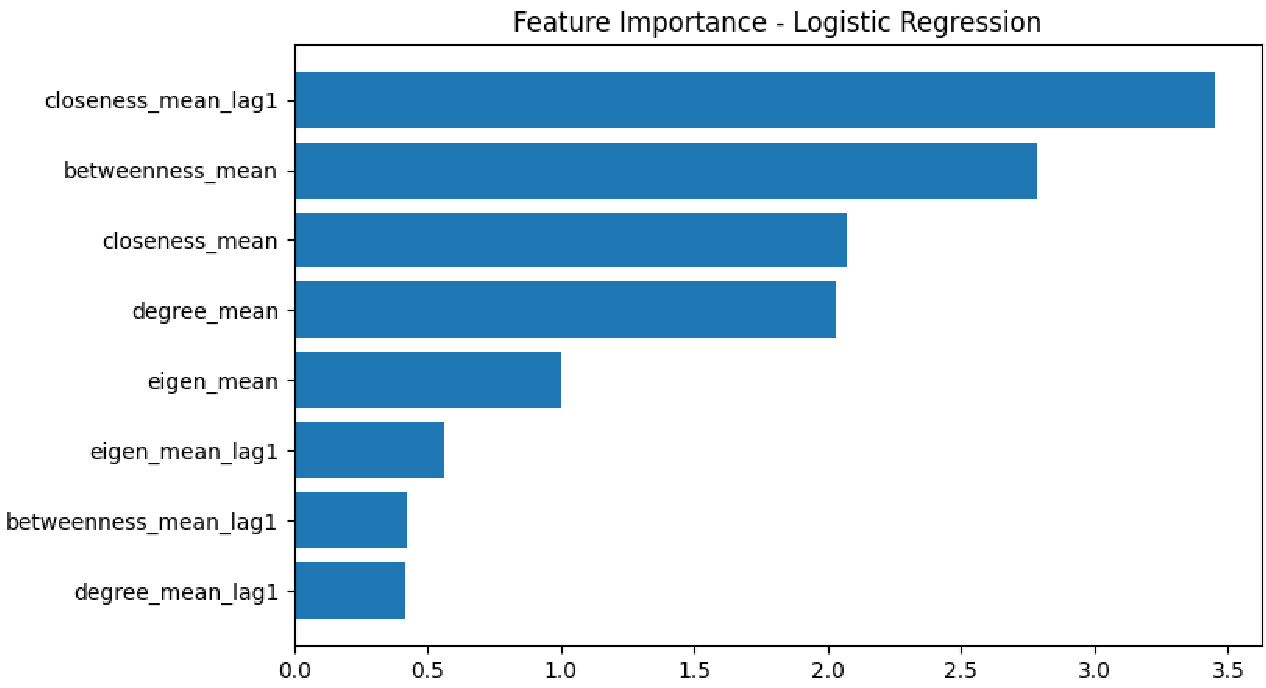

Figure C.7 : Method 1 feature importance for logistic regression (Co-Occurrence Network)

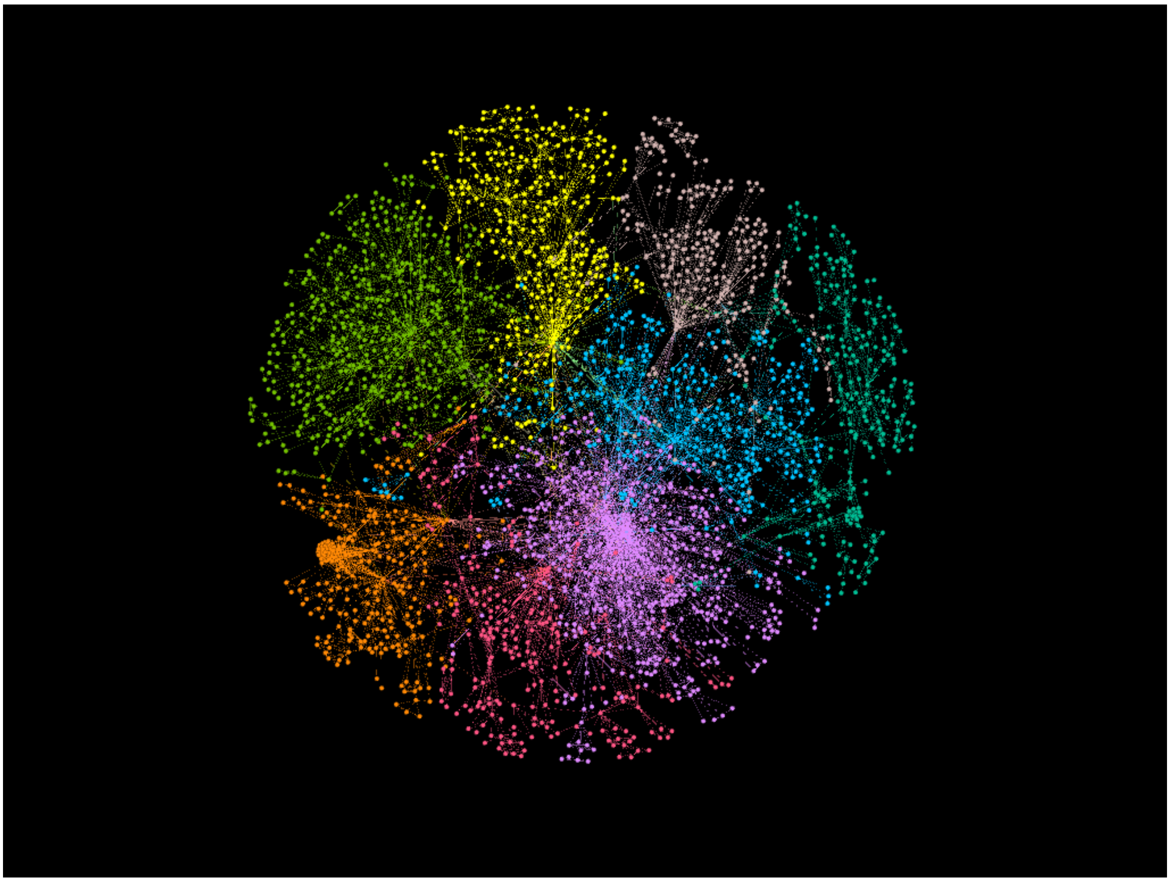

Figure C.8 : Visualizing the Co-Occurrence Network (for one day)

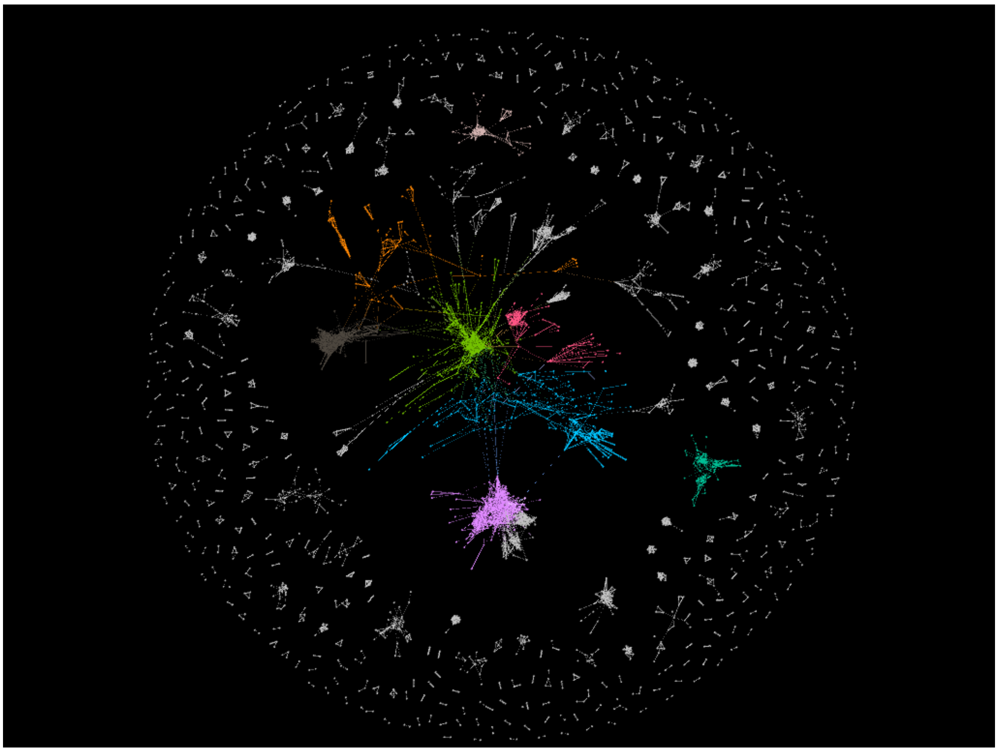

Figure C.9 : Visualizing the Embedding Network (for one day)

Appendix C : Additional Reading

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